IEEE VAST CHALLENGE 2023 · OCEANUS / FISHEYE INTERNATIONAL

Signal Hoist

Answer sheets for Mini-Challenges 1, 2 and 3

Live interactive site: amukherjee1991.github.io/Vast23
Repository: github.com/amukherjee1991/Vast23

Independent portfolio response written after the challenge; not an official 2023 entry. Every figure is a screenshot of an interactive view on the live site.

1. **Mini-Challenge 1: contextualising the FishEye knowledge graph**
2. **Mini-Challenge 2: trade patterns, predicted links and restarts**
3. **Mini-Challenge 3: anomalous business groups**

VAST Challenge 2023: Mini-Challenge 1

Entry Name: Signal Hoist

Team Member(s): [Name] — [Affiliation], GitHub @amukherjee1991

Tools Used: Python (pandas, NetworkX, scikit-learn, SciPy, PyArrow), React, D3, Vite, Playwright

Approximately how many hours were spent working on this submission in total? [AUTHOR TO FILL: hours]

May we post your submission in the Visual Analytics Benchmark Repository after VAST Challenge 2023 is complete? [AUTHOR TO FILL: Yes / No]

Video: Not provided

Interactive version: https://amukherjee1991.github.io/Vast23/#/mc1 (every figure below is a screenshot of a view on this page; the views are interactive there)

Questions

Figures are numbered *question.figure*: Figure 1.3 is the third figure in the answer to Question 1. Verdicts use one scale throughout: **Suspect**, **Possible**, **Insufficient evidence**, **Unlikely**, each with a confidence of **High**, **Medium** or **Low**.

The four entities named in FishEye's tip are Mar de la Vida OJSC, 979893388, Oceanfront Oasis Inc Carriers and 8327.

Question 1

> *Use visual analytics to dynamically display and explore context around the suspected entities listed above. What did you learn about each one? Can you connect them to illegal fishing? Provide evidence for or against the case that each entity is involved in illegal fishing and use visual analytics to express confidence in your conclusions. Limit your response to 600 words and 6 images.*

The key context is a layer of weak links. 99.2% of the 11,069 edge rows have a weight between 0.80 and 0.99. The other 88 rows, weighted 0.03 to 0.77, form one connected component of 79 nodes (Figures 1.1, 1.2). This is not chance:

- **Size.** In 1,000 random draws of 88 edges, the largest component averaged 10.6 nodes and never exceeded 29 ($p = 0.001$).
- **Ownership.** 66% of the layer's edges are ownership, against 18% of all edges ($p = 2e\text{-}23$). It holds two circular-ownership loops.
- **FishEye.** "FishEye International" has 24 edges, 19 of them in this layer, so the layer may encode FishEye's own reporting.
- **Suspects.** Three of the four suspects are inside the layer ($p = 5e\text{-}5$); the fourth is one hop away.

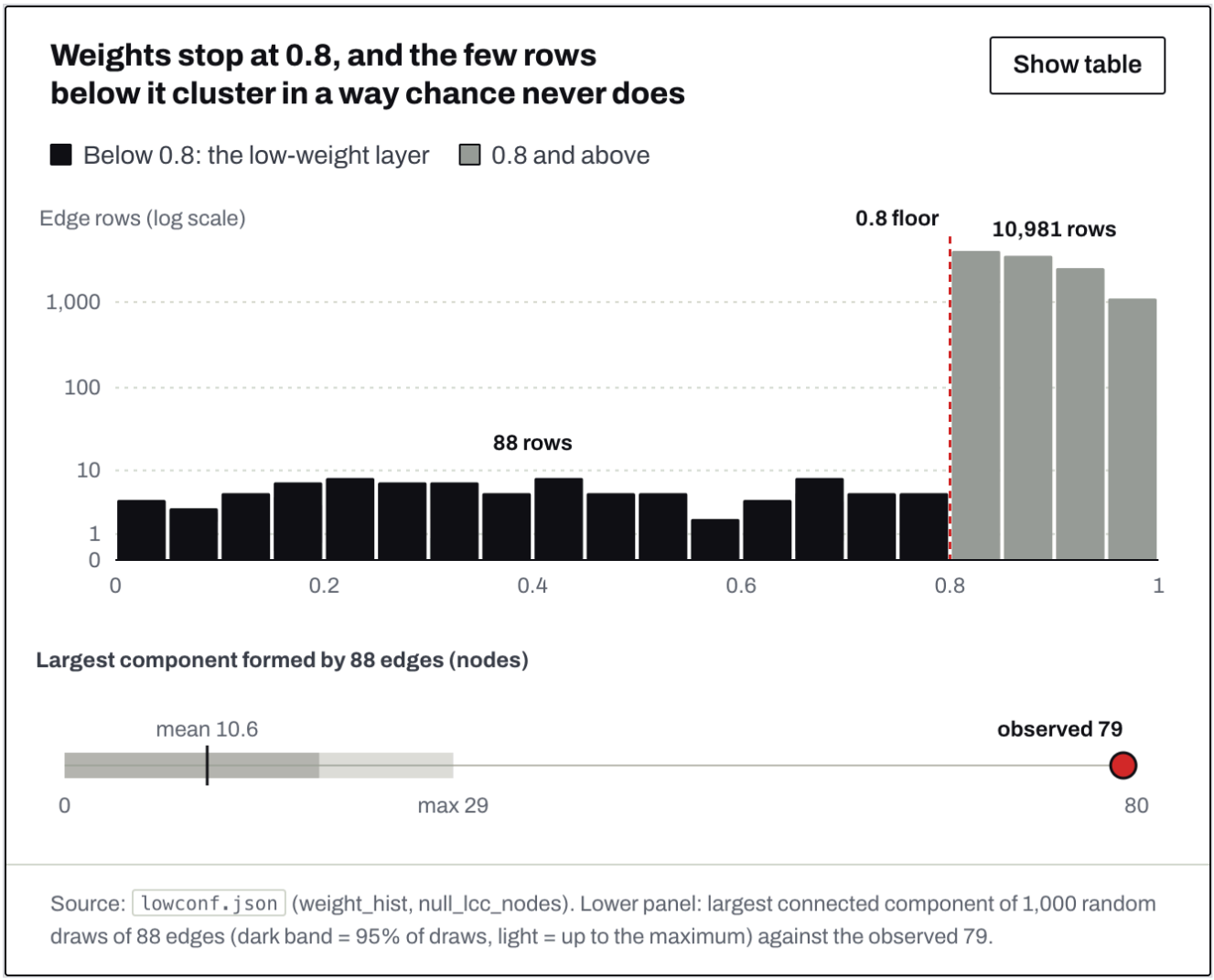


Figure 1.1. Weights stop at 0.8; the few rows below cluster as chance never does.

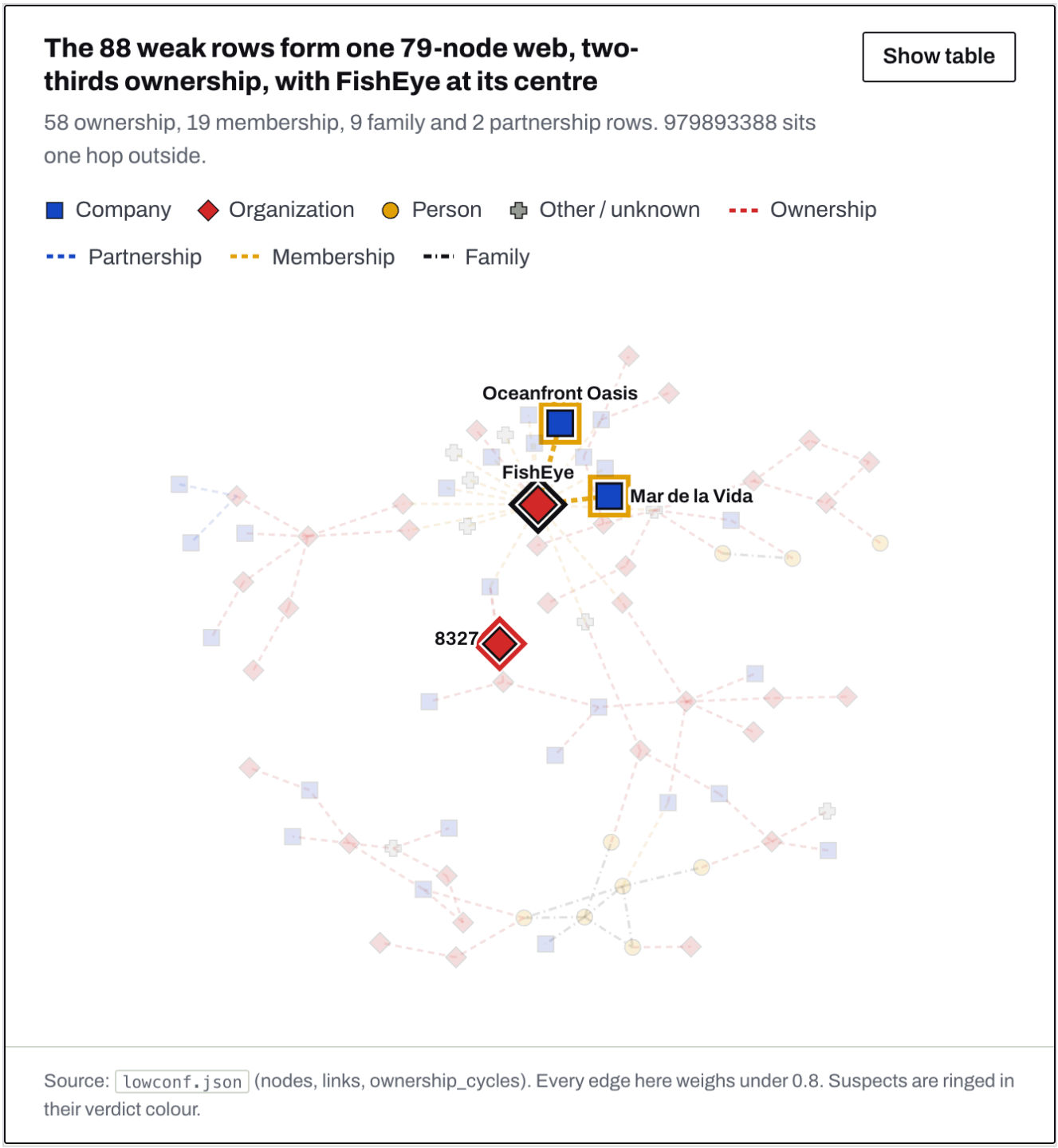


Figure 1.2. The 88 weak rows: one 79-node web, two-thirds ownership, FishEye at its centre.

Verdicts come from a published seven-signal rubric (4+ points Suspect, 2.5–3.5 Possible). Confidence counts full-point signals resting on edges weighted ≥ 0.8 or on structure alone (4+ High, 2–3 Medium, 0–1 Low). Ego views (Figures 1.3–1.6) ring each suspect in its verdict colour and dash weak edges.

8327: Suspect, Low confidence (5.0 points; Figure 1.3). A nameless organization with 13 neighbours. "Seabreeze Foods Ltd. Corporation United" owns it through one of the ten weakest edges in the graph (w = 0.12), and FishEye tags Seabreeze (w = 0.13). 8327 owns #40213337, which has six low-weight ownership rows and owns Sea Harvest, Sea Products and #58623386. 8327 has a membership edge (w = 0.82) to the garbled fragment "Illegal". A directed ownership chain runs from 8327 through #40213337, Sea Harvest, Amanda Mckenzie and Mar del Oeste Pic to 979893388, then to Oceanfront Oasis: six hops, weakest edge 0.32. Strongest structural signature of the four, but nearly every signal rests on edges below 0.8.

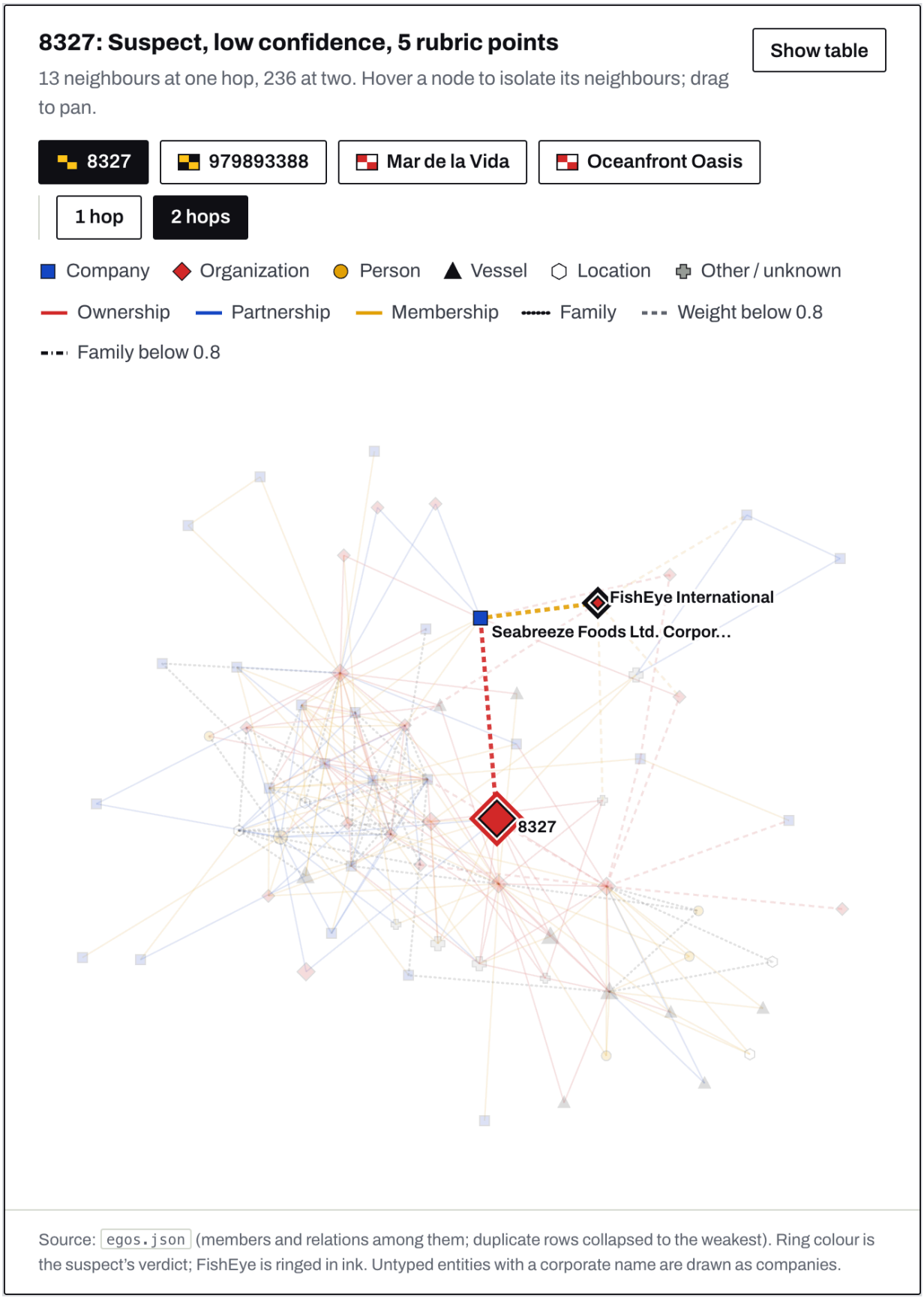


Figure 1.3. 8327: owned by Seabreeze through a 0.12 edge; FishEye tags Seabreeze.

979893388: Suspect, Medium confidence (4.0 points; Figure 1.4). A nameless hub: degree 177 and betweenness 0.096, both at the 99.8th percentile. It is one of 10 hubs forming a dense core (density 0.84). It owns 10 entities and is owned by 12; its upstream ownership tree reaches 307 entities within six hops. It owns Oceanfront Oasis through a high-weight edge (0.84), and its two-hop neighbourhood holds the fragments "ipoa-iuu" and "poached". It is not in the layer but sits next to two layer hubs. It reads as a holding layer, or an entity the extractor over-linked.

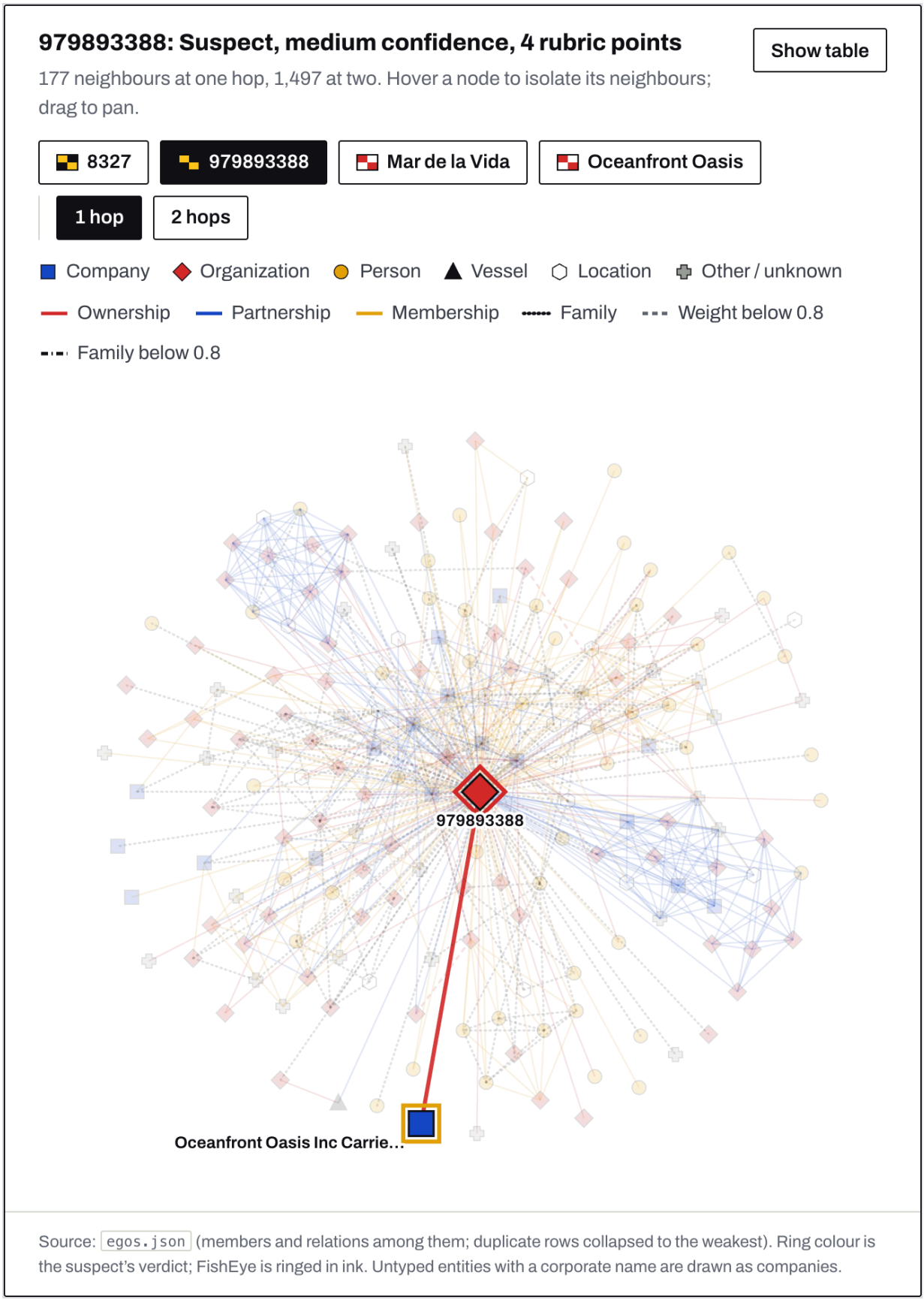


Figure 1.4. 979893388, a hub, owns Oceanfront Oasis through a strong 0.84 edge.

Mar de la Vida OJSC: Possible, Low confidence (3.0 points; Figure 1.5). Untyped, 15 neighbours, mostly memberships, 6 of them people. Its only low-weight edge is a FishEye membership (0.46). It owns one person, Jon White, and has no ownership chain. At two hops it reaches "Officer Pleads Guilty" and "Special Action Group Combat Organized Crime"; betweenness is at the 95th percentile. The weak FishEye tag is the main evidence.

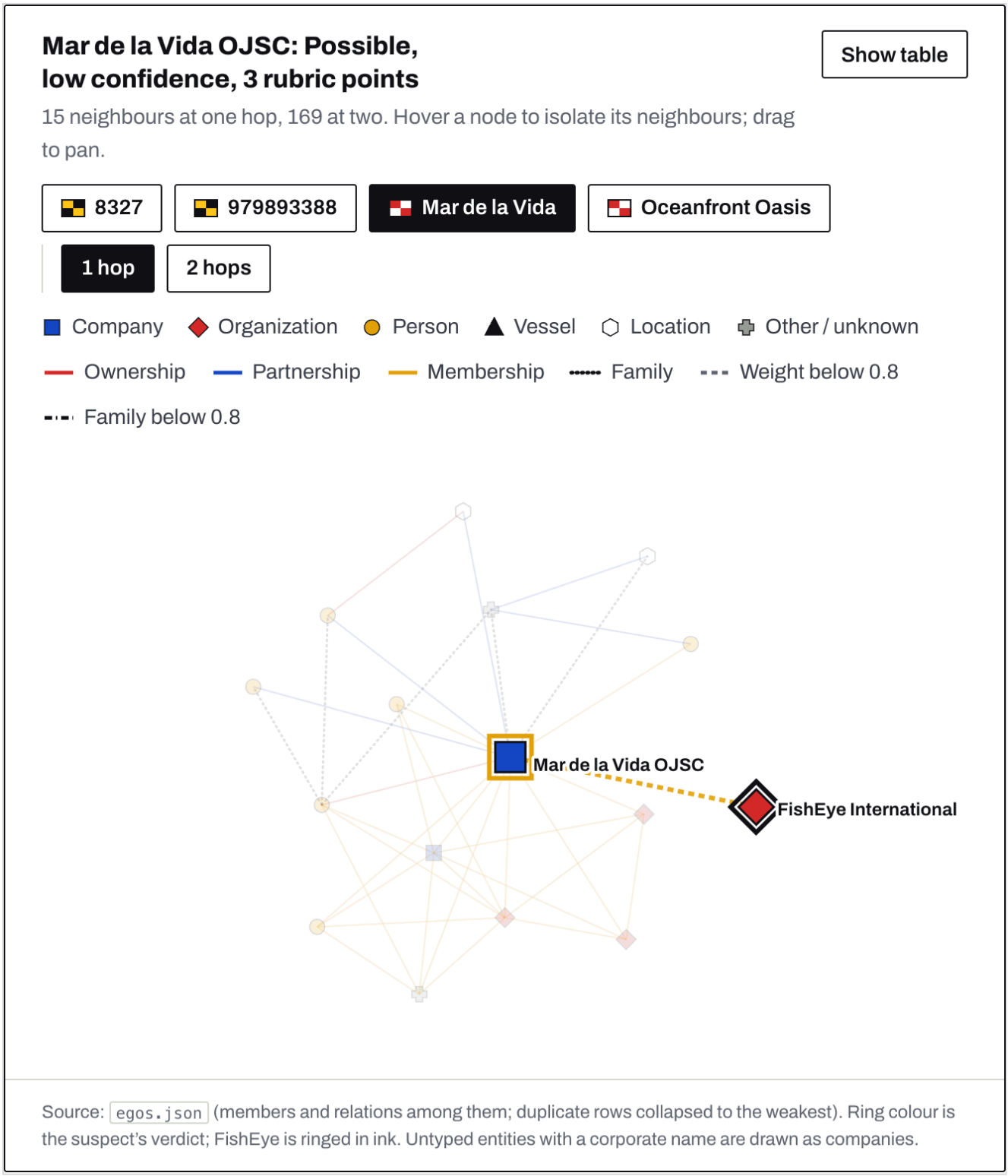


Figure 1.5. Mar de la Vida touches the layer only through a weak FishEye membership.

Oceanfront Oasis Inc Carriers: Possible, Low confidence (3.0 points; Figure 1.6). Four edges, all incoming: a FishEye membership (0.45), ownership by 979893388 (0.84), and two family edges from nameless organizations, probably noise. A thin subsidiary of a suspect.

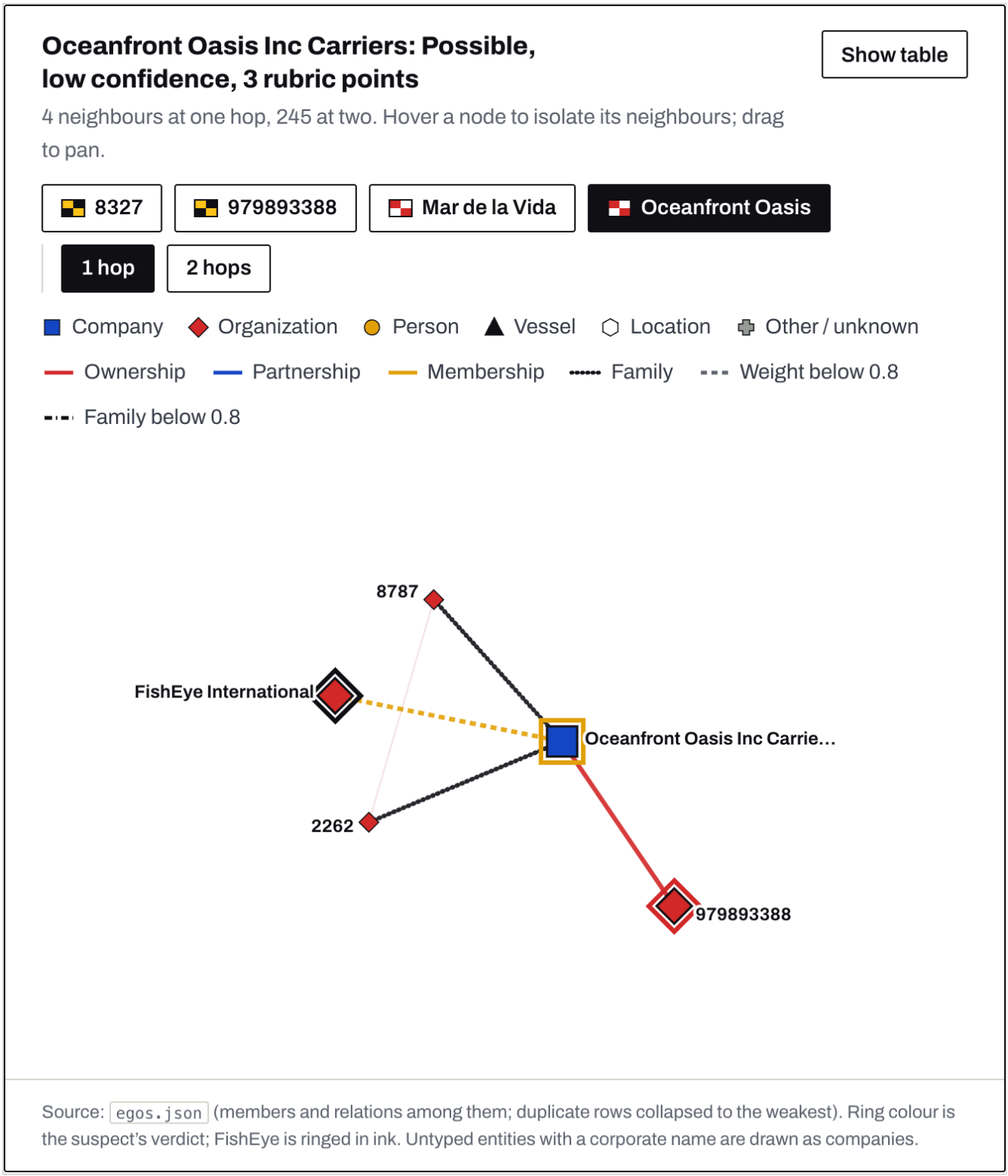


Figure 1.6. Oceanfront Oasis: four incoming edges and nothing outgoing.

Caveats. The data notes define neither `weight` nor ownership direction, so reading weight as extraction confidence is an assumption. 356 nodes (10%) are NLP debris.

Question 2

> *Use your visual analytics tool to compare and contrast what you learned about the suspect entities. Are there patterns that may indicate illegal activity? Use visual analytics to express confidence that a pattern exists and where uncertainty may be affecting this confidence. Limit your response to 400 words and 4 images.*

The suspects fall into two pairs (Figure 2.1).

Tagged small entities: Mar de la Vida and Oceanfront Oasis. Named, untyped, with 15 and 4 neighbours. Each touches the layer only through one weak FishEye membership (0.46, 0.45), and FishEye is their only shared neighbour.

Ownership intermediaries: 8327 and 979893388. Nameless numeric organizations that both own and are owned (3 and 2; 10 and 12). Only they fire the rubric's shell marker. They differ in scale: 979893388 is at the 99.8th percentile of degree, 8327 at the 91st.

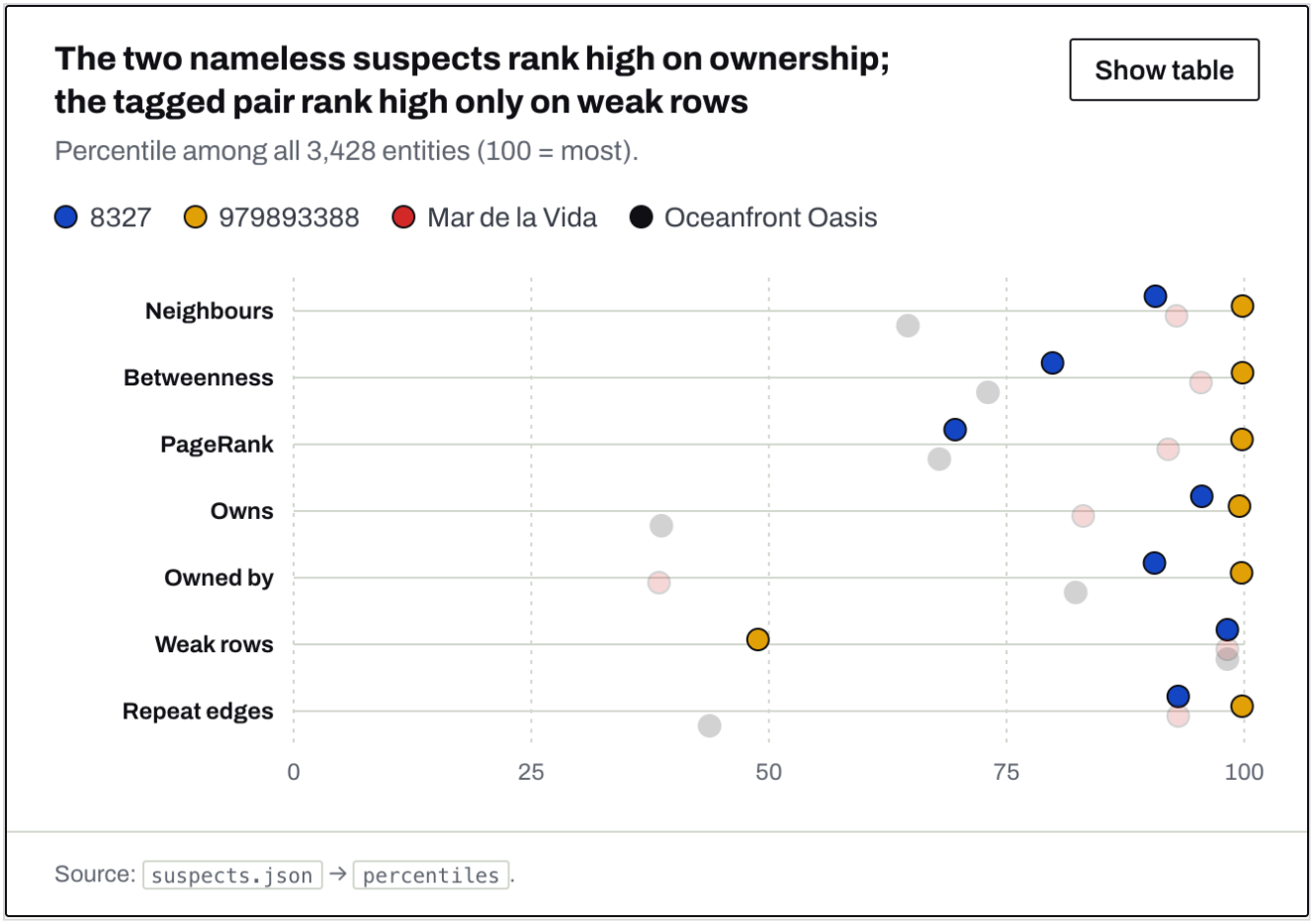


Figure 2.1. The nameless pair rank high on ownership; the tagged pair rank high only on weak rows.

Patterns that point to illegal activity:

1. **Ownership in the weak layer.** Two-thirds of its edges are ownership, with two circular loops (4 and 5 entities) and a five-hop directed chain.
2. **One chain through three suspects.** The six-hop chain links 8327, 979893388 and Oceanfront Oasis (Figure 2.3). Mar de la Vida has no ownership path to another suspect.
3. **FishEye tags.** Two suspects are tagged directly; 8327 through its owner, 979893388 through a neighbour.
4. **Nameless intermediaries.** 34 of the layer's 79 nodes are nameless numeric organizations.

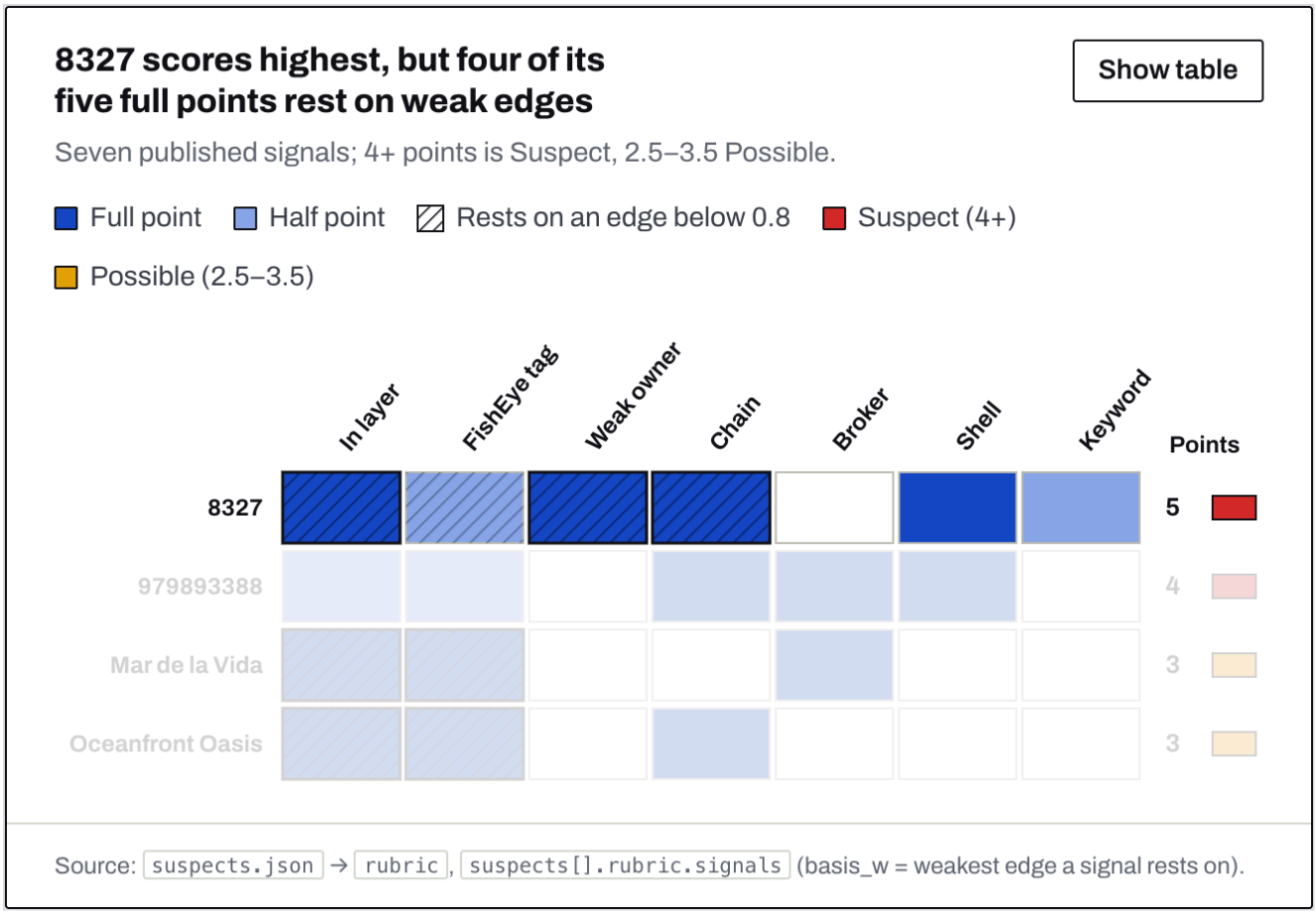


Figure 2.2. 8327 scores highest, but four of its five full points rest on weak edges.

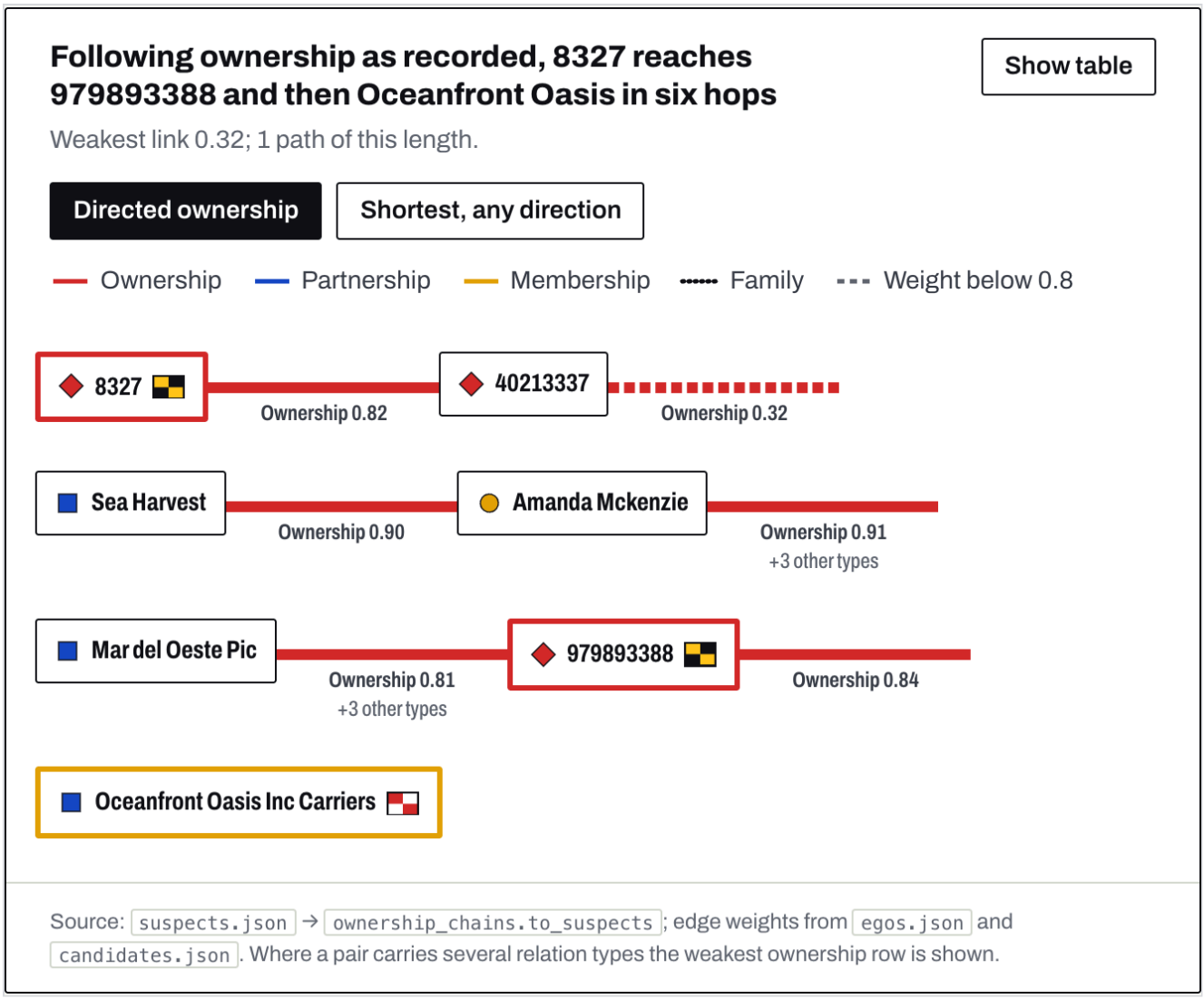


Figure 2.3. Following ownership as recorded, 8327 reaches 979893388 and then Oceanfront Oasis in six hops.

Uncertainty, and how the views show it:

- **Distances.** The suspects sit 2.17 hops apart on average, against 3.84 for random pairs, but 8% of random degree-matched sets are as close (Figure 2.4). Their closeness is mostly what their degrees predict. Pattern confidence: Low.
- **Communities.** Louvain places the suspects in three communities (modularity 0.69), and across five seeds only 0.33–0.63 of each suspect's community neighbours stay with it. No stable shared community.
- **Ownership direction.** Read literally, the chain has the company Sea Harvest owning the person Amanda Mckenzie. Direction cannot be trusted.
- **Edge weights.** The two Suspect verdicts differ in confidence: 979893388 rests on high-weight edges and structure (Medium), 8327 on edges from 0.12 to 0.32 (Low; Figure 2.2).

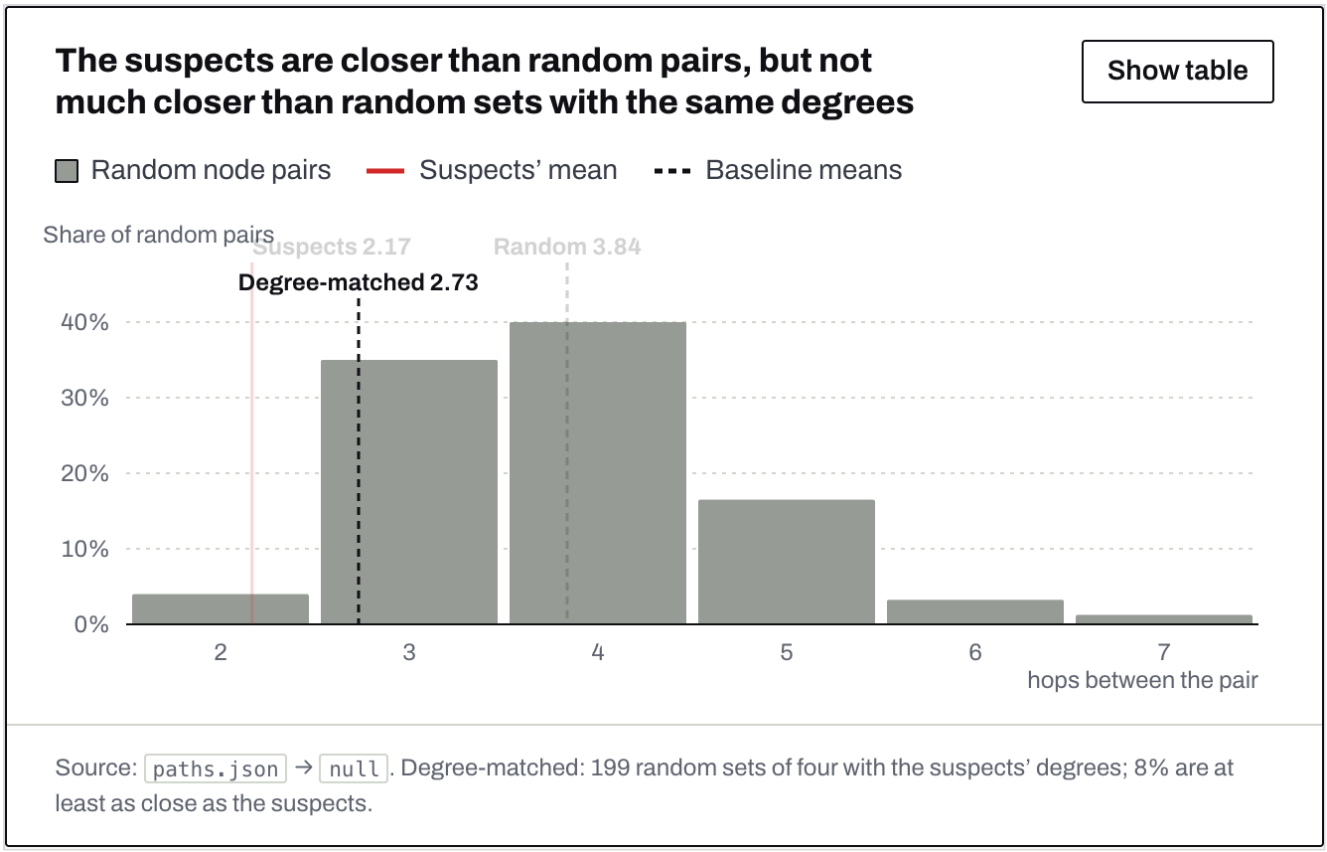


Figure 2.4. Suspects are closer than random pairs, but not much closer than random sets with the same degrees.

Overall confidence that a pattern exists: **Medium** for the weak ownership layer, **Low** for any tie between specific suspects.

Question 3

> *What other companies should FishEye investigate for illegal fishing? Show how your visual analytics can be used to find entities that are worthy of further investigation. Limit your response to 600 words and 6 images.*

How the tool surfaces them. It scores all 1,209 companies and organizations on seven structural features, each converted to a percentile before weighting: low-weight edges (0.25), proximity to suspects (0.20), shared neighbours (0.15), ownership counterparties (0.15), relation-type profile similarity (0.10), repeated or mixed-type edges (0.10) and a FishEye tag (0.05). Each candidate comes with its feature contributions (Figure 3.1), an evidence subgraph of at most 22 nodes, and the verdict rubric used on the suspects (Figures 3.4, 3.5).

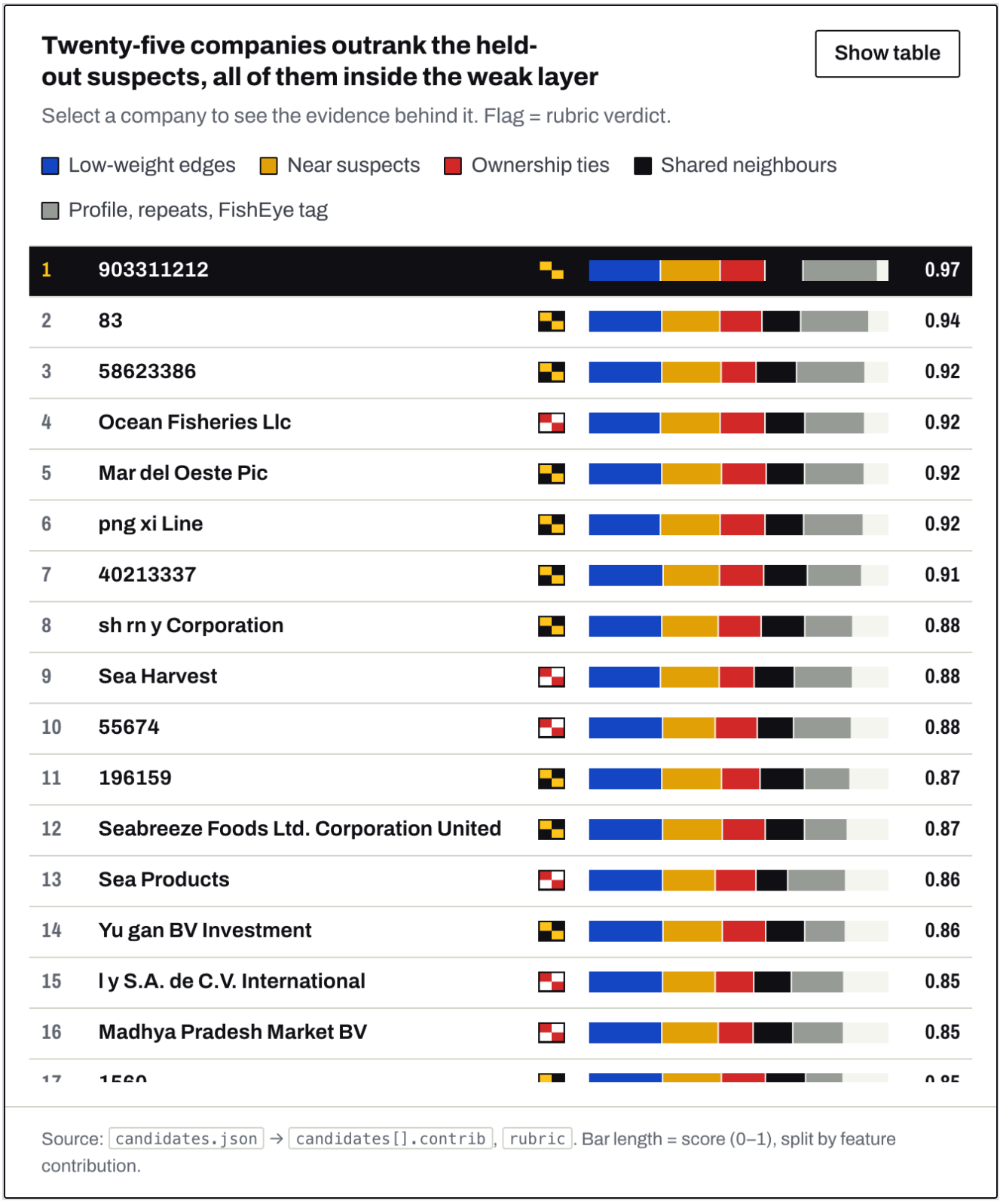


Figure 3.1. The ranked candidate list; bar segments show each feature's contribution.

Validation.

- **Held-out suspects.** Holding each suspect out and seeding with the other three, the held-out one ranks 11th, 51st, 41st and 35th of 1,210 (top 1–4%). Ranking by degree places them 96th, 7th, 528th and 128th (Figure 3.2).
- **Weight changes.** Across 300 random reweightings, the ranking keeps a mean Spearman correlation of 0.98 (minimum 0.89).
- **Ablation.** Without the low-weight feature only 8 of the top 25 stay (Figure 3.3). The list is mainly a way to rank the layer.

All 25 top candidates sit in the weak layer; 4 are hubs.

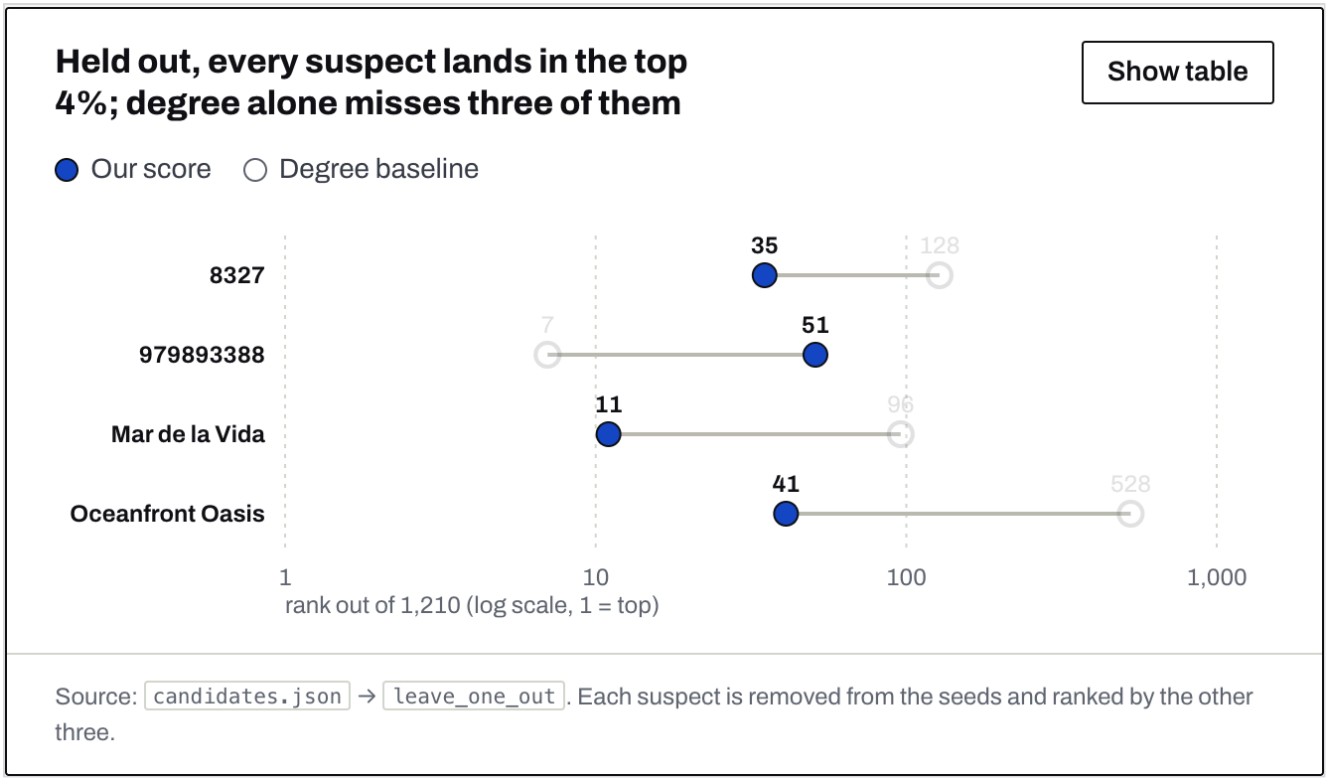


Figure 3.2. Held out, every suspect lands in the top 4%; degree alone misses three of them.

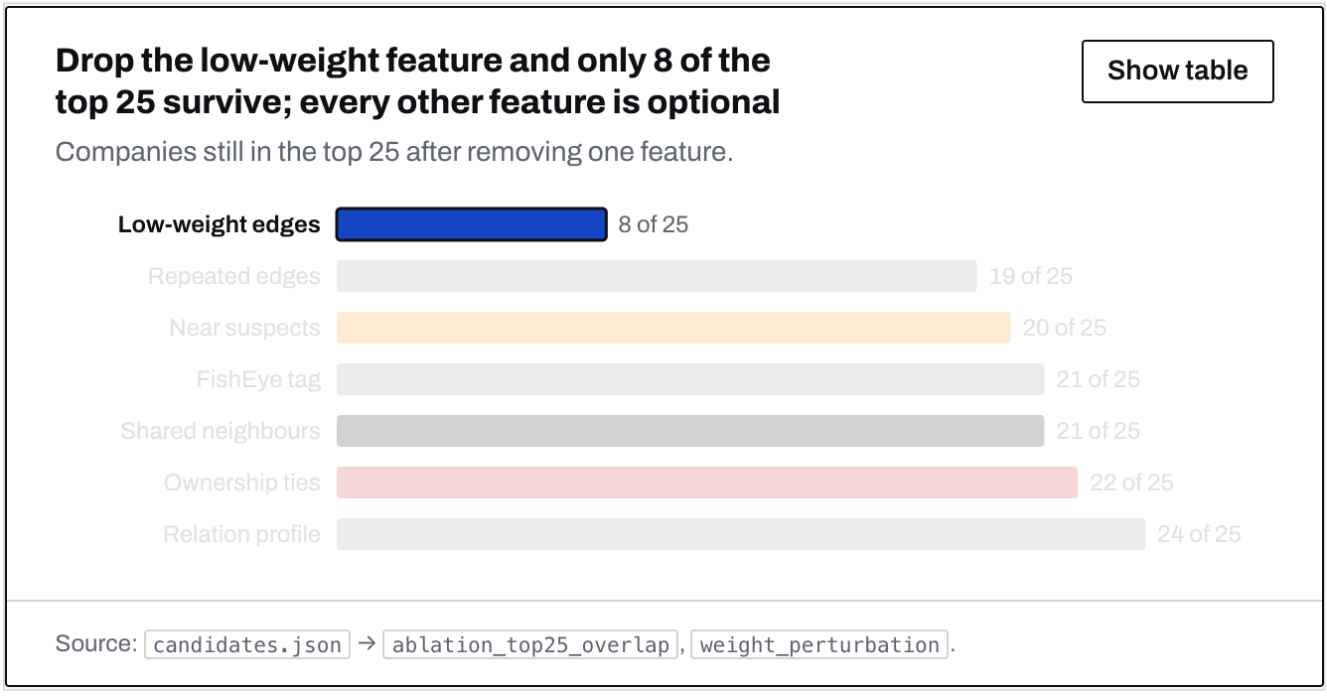


Figure 3.3. Drop the low-weight feature and only 8 of the top 25 survive.

Investigate first: rubric Suspect with Medium or High confidence.

- **#903311212** (rank 1, score 0.97; Figure 3.4). A hub with 234 neighbours, 43 owned and 42 owners. FishEye tags it, it has a low-weight ownership edge, it links to the "poached" fragment, and it reaches 979893388 through Mar del Oeste Pic.
- **#48** (rank 18, the only **High**-confidence result; Figure 3.5). FishEye tags it with a high-weight edge (0.95). It owns 5 and is owned by 4; an ownership chain reaches 979893388 through png xi Line and Ocean Fisheries Llc.

- **#40213337** (rank 7). The holding 8327 owns, with six low-weight ownership rows; it owns Sea Harvest, Sea Products and #196159.
- **#196159** (rank 11). Owned through #40213337 and next to a FishEye-tagged entity.
- **#83** (rank 2). FishEye-tagged; owns 4, owned by 1, with a chain to 979893388.
- **Mar del Oeste Pic** (rank 5), **png xi Line** (rank 6), **sh rn y Corporation** (rank 8). The first two are hub-core members that own or connect to 979893388; the third has six ownership ties and shares neighbours with the suspects.

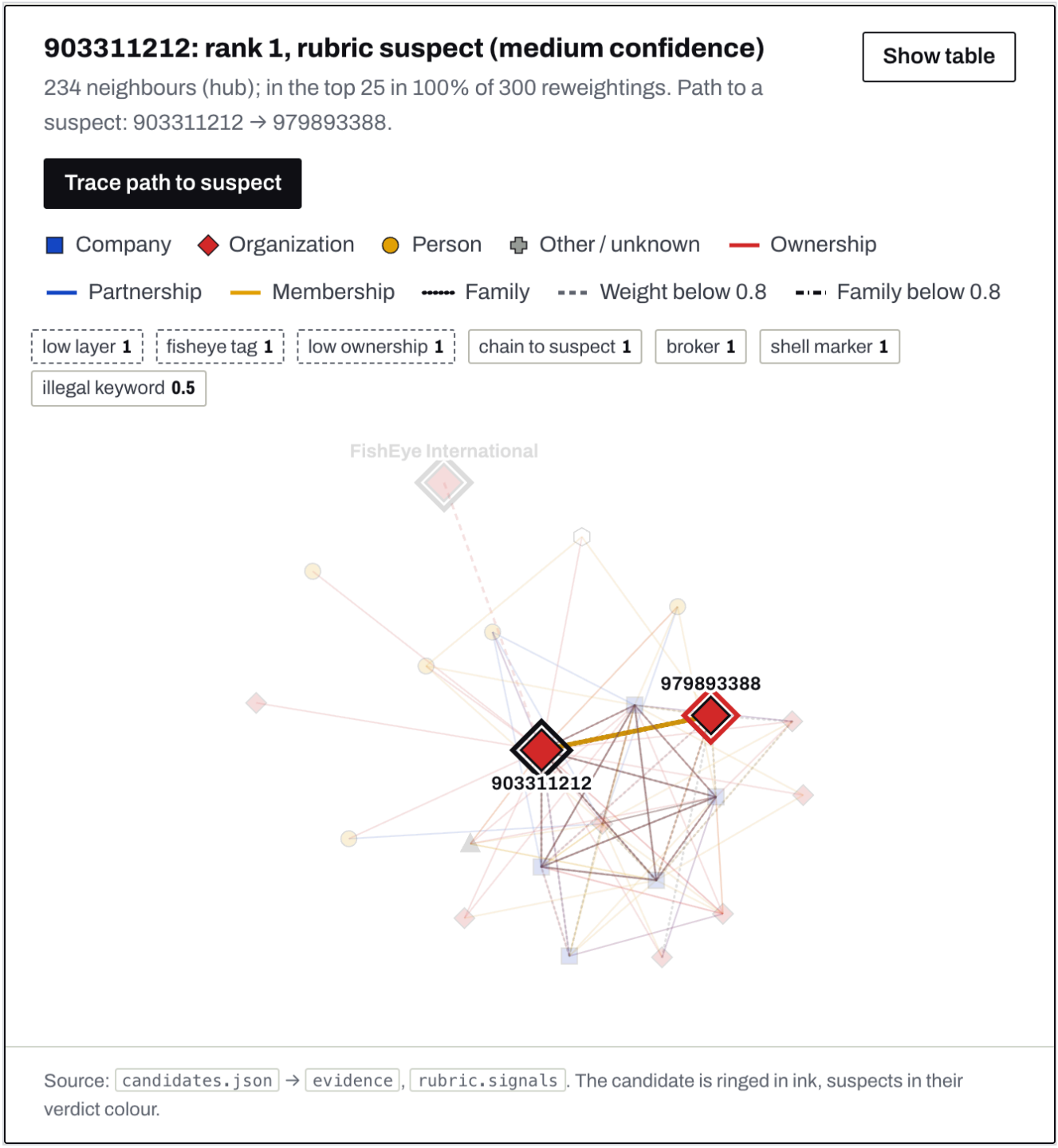


Figure 3.4. Evidence subgraph for #903311212, rank 1, rubric Suspect (Medium).

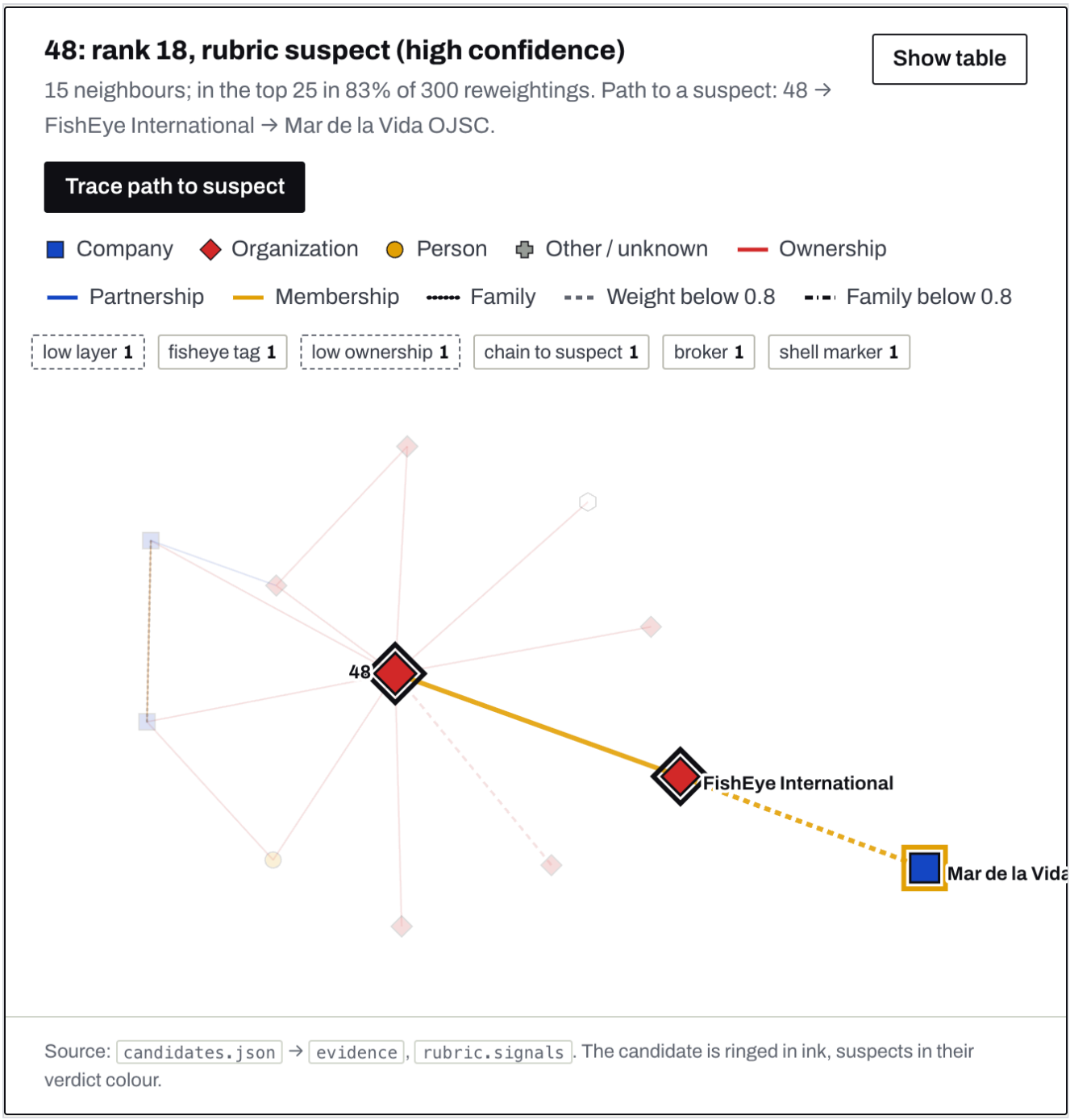


Figure 3.5. Evidence for #48, the only High-confidence candidate, with its shortest path to a suspect traced.

Next tier: Suspect with Low confidence, or Possible. Seabreeze Foods Ltd. Corporation United (8327's FishEye-tagged owner); Yu gan BV Investment; #1560, #8814 and #923825740; Ocean Fisheries Llc, Sea Harvest, Sea Products and #55674.

Structures, not just entities.

- Two circular-ownership loops: #21538 → #52193 → #958085 → #55674 → #21538, and American → #5421 → #81 → "l y S.A. de C.V. International" → #59 → American.
- A five-hop chain: Darin Torres → Federated → #1354 → Ht Shipping → Yu gan BV Investment → #4942.
- Nine people in the layer, including Daniel Henderson, Traci Shields and Joseph Smith. Weak family edges tie them together and several own layer companies, so they may be the beneficial owners.

Confidence. Medium for the layer as a whole, which is statistically clear. **Low to Medium** for each individual company, because the strongest edges in that evidence are also the least reliable ones.

Question 4

> *Reflection: What was the most difficult aspect of working with this knowledge graph? Did you have the tools and resources you needed to complete the challenge? What additional resources would have helped you? Limit your response to 300 words*

The hardest part was deciding which parts of the graph to trust. The extraction errors are not random; they form structure that can look like evidence:

- About 10% of nodes are NLP debris (garbled text, words like "weeks" labelled as vessels), and 605 nodes have no type.
- Family edges join two non-people 46% of the time.
- 322 edge rows repeat the same source, target and type; 124 pairs carry several relation types at once.
- A 10-node hub core holds 3.4% of all edge rows; Ocean Fisheries Llc and SeaSpray Wave SRL Solutions share 40 edges of all four types.
- Integer and string ids collide (77 vs "77", 11 cases), so every id needs a type-aware key.
- The data notes declare 3,721 nodes and 7,422 edges; the file has 3,428 nodes and 11,069 rows, and one row misspells its weight field.

The most important signal, edge weight, is undocumented, and so is ownership direction. So the weak layer was treated as a hypothesis: tested against a random-edge baseline, checked for predicting held-out suspects better than degree, and every verdict records the weight of the edges it rests on. Hubs also hide structure, because many shortest paths run through them and community detection is unstable; the tool compares paths with and without hubs and runs Louvain with several seeds.

Tools and resources. Open-source Python and D3 were sufficient. What would have helped most is a data dictionary defining `weight` and the direction of ownership edges, and ground truth for even a few entities to calibrate the rubric.

VAST Challenge 2023: Mini-Challenge 2

Entry Name: Signal Hoist

Team Member(s): [Name] — [Affiliation], GitHub @amukherjee1991

Tools Used: Python (pandas, NetworkX, scikit-learn, SciPy, PyArrow), React, D3, Vite, Playwright

Approximately how many hours were spent working on this submission in total? [AUTHOR TO FILL: hours]

May we post your submission in the Visual Analytics Benchmark Repository after VAST Challenge 2023 is complete? [AUTHOR TO FILL: Yes / No]

Video: Not provided

Interactive version: https://amukherjee1991.github.io/Vast23/#/mc2 (every figure below is a screenshot of a view on this page; the views are interactive there)

Questions

Figures are numbered *question.figure*: Figure 2.3 is the third figure in the answer to Question 2. Confidence is stated as **High**, **Medium** or **Low**.

Data preparation. The graph has 5,464,378 shipment edges; 155,291 (2.8%) exactly duplicate another and are dropped, leaving 5,309,087 across 34,576 companies, 245,807 shipper→receiver pairs and 84 months (2028-01 to 2034-12). HS codes for chapters 01–09 lost their leading zero (306170 is 0306.17, frozen shrimp), and the bundles store them with five digits (30729); appending a "0" matches the graph's codes 100% of the time. "Fish" means HS headings 0301–0309 plus 1604/1605: 13.9% of edges.

Question 1

> *Use visual analytics to identify temporal patterns for individual entities and between entities in the knowledge graph FishEye created from trade records. Categorize the types of business relationship patterns you find. Limit your response to 600 words and 6 images.*

The market as a whole (Figure 1.1). Yearly volume peaks in 2030 at 831k shipments and falls to 688k in 2034. Fish holds steady at about 96k–116k a year, so its share rises from 12.7% (2028) to 15.6% (2034). Fish is mildly seasonal: a calendar index of 0.84–0.89 in February–April and 1.05–1.08 in July–December, against 0.91–1.09 for all goods.

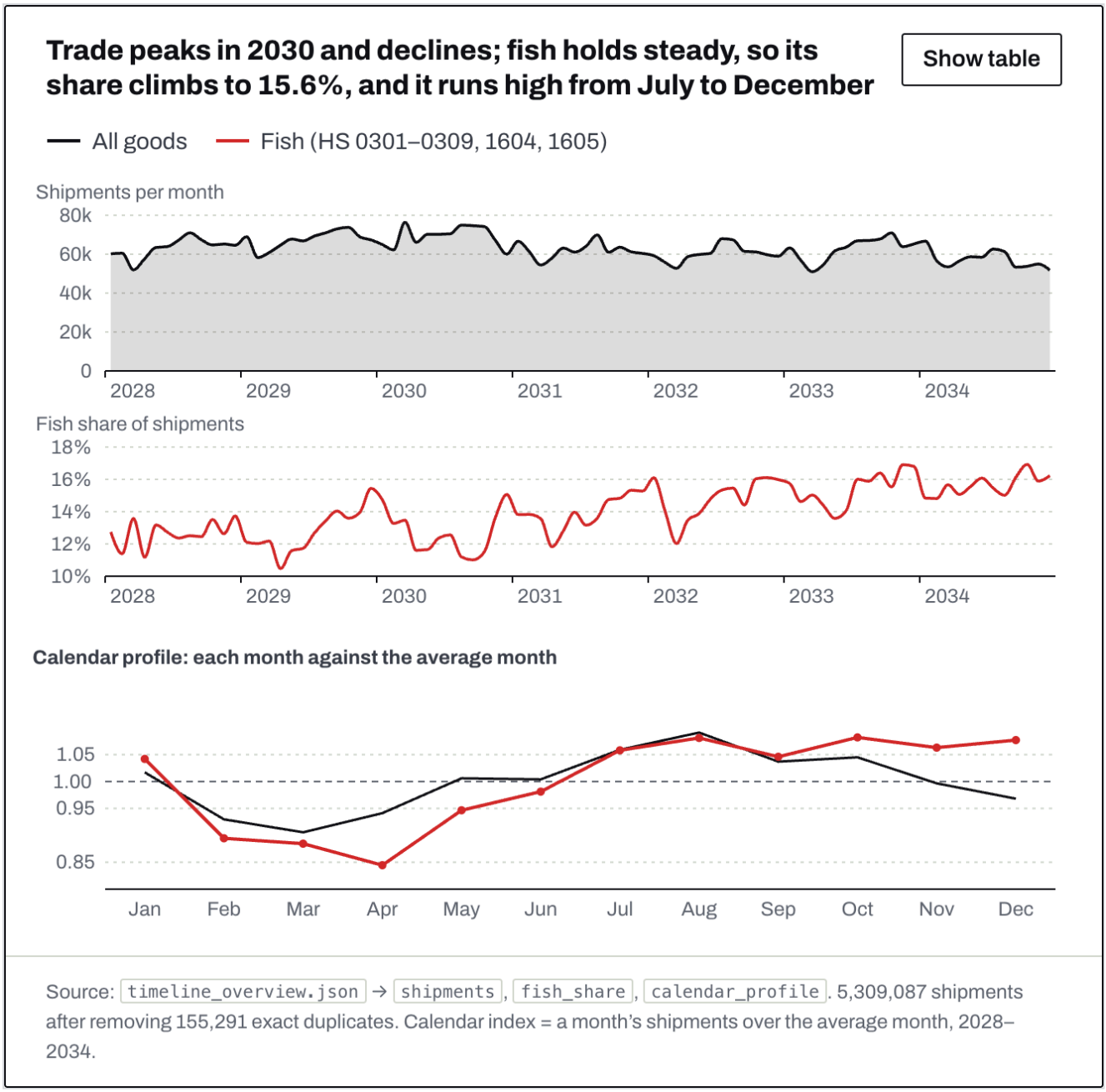


Figure 1.1. Trade peaks in 2030 and declines; fish holds steady, so its share climbs to 15.6%.

Individual companies (Figure 1.2). The two roles behave very differently:

- **Shippers are durable.** Median span 59 months; 4,875 of 12,217 are active for 73–84 months.
- **Receivers are fleeting.** Median span 3 months; 44% appear in a single month.
- **Fish shippers churn more.** 2,139 of 4,299 fish-majority shippers are *transient* (they appear after the data starts and disappear before it ends), against 4,740 of 12,217 shippers overall.

The 150 per-company series online show flat hubs, step-ups, step-downs and bursts.

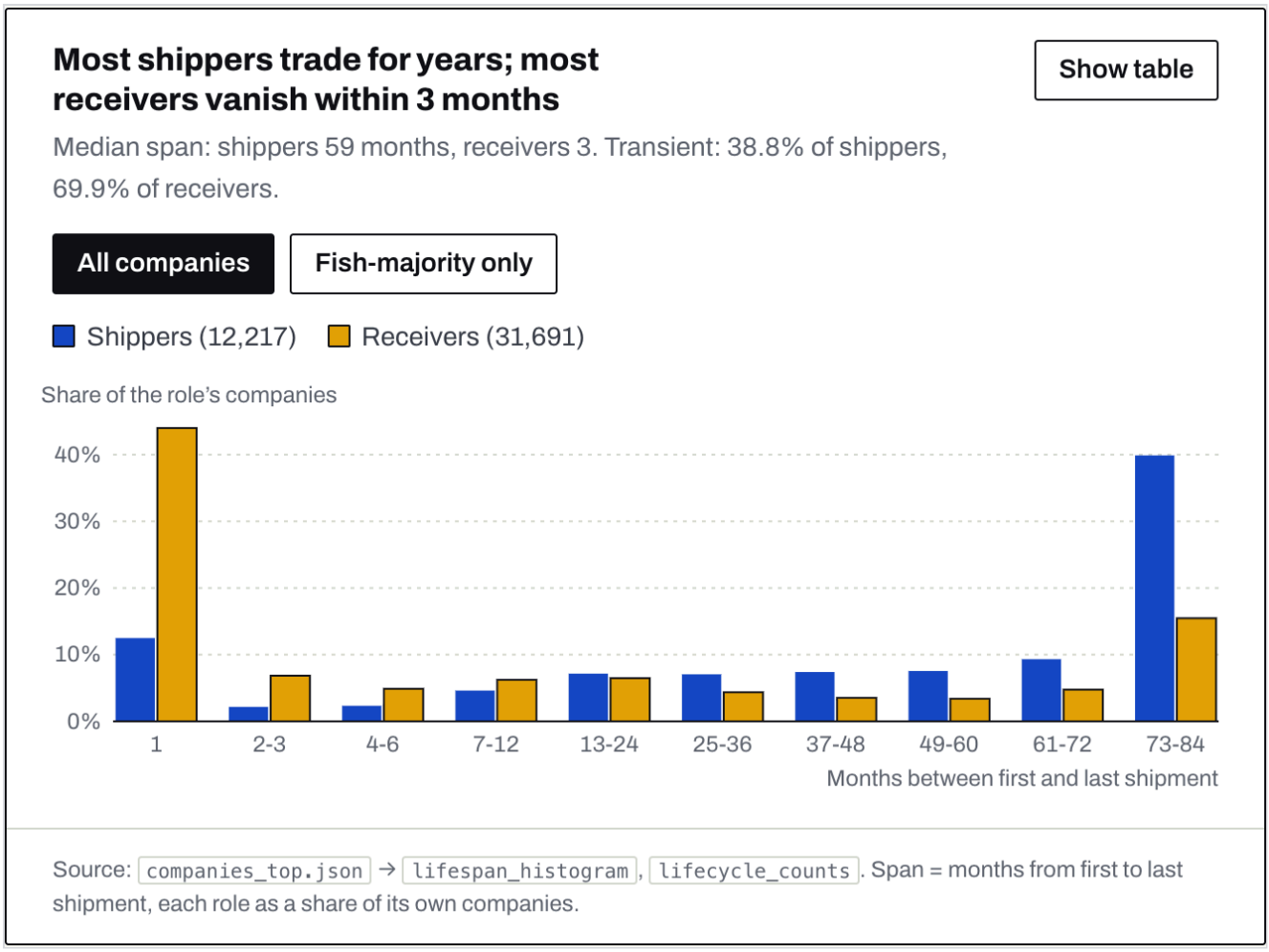


Figure 1.2. Most shippers trade for years; most receivers vanish within 3 months.

Relationships between companies (Figure 1.3). Each pair's monthly series is classified by span, coverage, trend, seasonality and how strongly it ends; rules apply in order:

Pattern	Pairs	Share of shipments	Signature
One-off	108,067 (44%)	2.6%	a single month
Intermittent	60,968	14.5%	recurring but irregular
Short burst	31,604	3.0%	≤ 6 months, then nothing
Fading	16,729	21.9%	second half ≤ ½ of the first
Ramping	13,385	16.0%	second half ≥ 2× the first
Abrupt stop	5,994	8.3%	regular trade that stops at full strength ≥ 12 months before the data ends
Steady	4,634 (1.9%)	32.6%	≥ 24 months, ≥ 60% of months active
Seasonal	4,426	1.2%	≥ 75% of trade in the same 3 calendar months

The network is a small core of steady pairs carrying a third of all shipments, ringed by a large fringe of one-off links. Only 737 pairs trade in both directions, so relationships run one way. Fish pairs follow almost the same mix; abrupt stops are slightly *rarer* among them (1.7% vs 2.4%), so an abrupt stop alone is not a sign of illegal fishing. As a cross-check, k-means (k = 8) on 101,115 recurring pairs agrees with the rules at a weighted purity of 0.57: steady, fading and ramping separate cleanly; the catch-all "intermittent" does not.

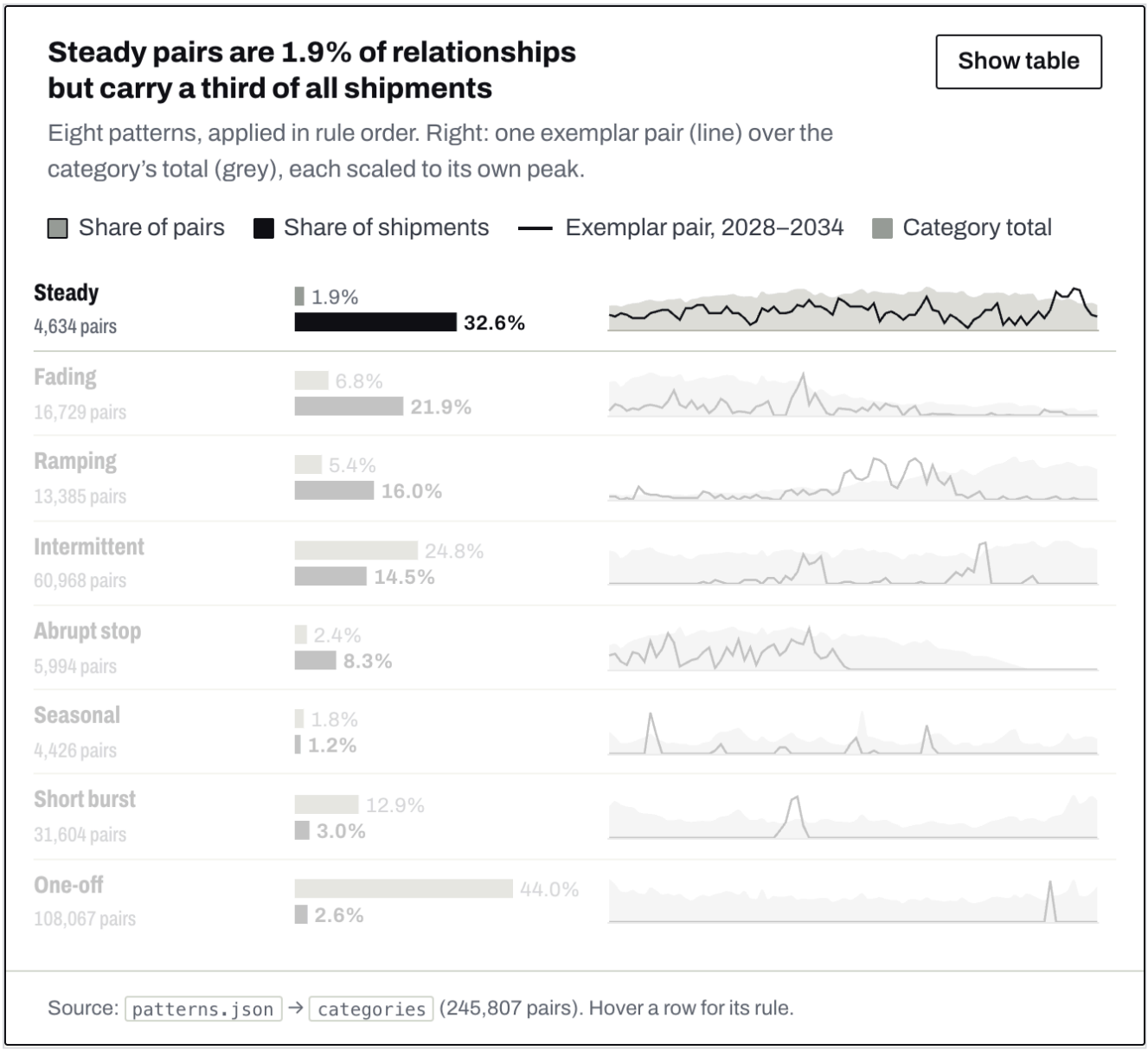


Figure 1.3. Steady pairs are 1.9% of relationships but carry a third of all shipments.

Between entities: successors (Figure 1.4). We took company pairs with very similar partner and product profiles (partner and HS cosine ≥ 0.7, ≥ 3 shared partners) and counted them by the gap between the earlier company's last month and the later one's first month:

- **Twins.** Most such pairs *coexisted* for 12–36 months (31 shipper pairs fall in −36..−6). These are sister firms, not replacements.
- **Restarts.** A second, narrower peak sits at gaps of 0–6 months: 1.7 pairs per gap-month, against 0.3 in the −6..0 window just before.

That excess is the signature of a company stopping and a look-alike starting straight afterwards: real but small, so successor pairs are rare and specific (see Question 4).

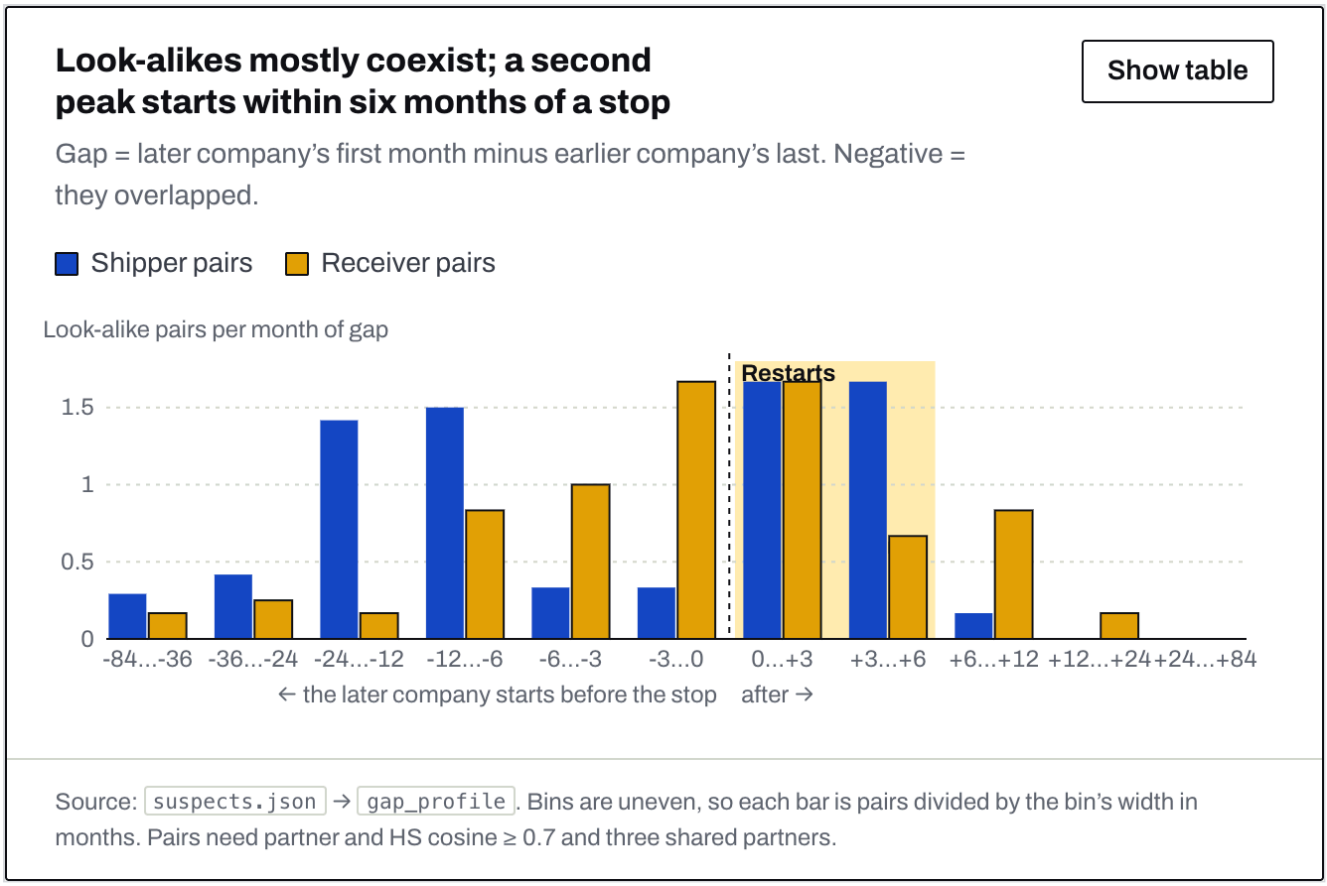


Figure 1.4. Look-alikes mostly coexist; a second peak starts within six months of a stop.

Confidence: High for the aggregate patterns, which are counts over the full graph. The category boundaries are analyst choices, documented with each category in the tool.

Question 2

> *Evaluate the sets of predicted knowledge graph links FishEye has provided using visual analytics. Which sets are most reliable for completing the graph? Limit your response to 600 words and 6 images.*

Each bundle is scored against the real graph on 11 components, each 0–1: repetition, novelty, schema, dates, month mix, pair and product history, HS mix, concentration, weight, value per kg, and node countries (Figure 2.1). Distribution tests compare each bundle with **200 same-size random samples of real 2034 edges**, so small bundles are not punished for noise. Gates override the score:

- **Critical gates force "Unreliable":** under 25% of links distinct, fewer than 20 distinct links, or over 95% exact copies of graph edges.
- **Major gates cap the verdict at "Questionable":** under 50% of links carry a weight, or over 5% fall outside the graph's date window.

Novelty is two-sided. An exact match with a real edge (same shipper, receiver, HS code and day) is almost impossible by chance: some matches anchor a bundle to real records; all matches mean it adds nothing. The five diverse bundles independently converge on 42–59% matches.

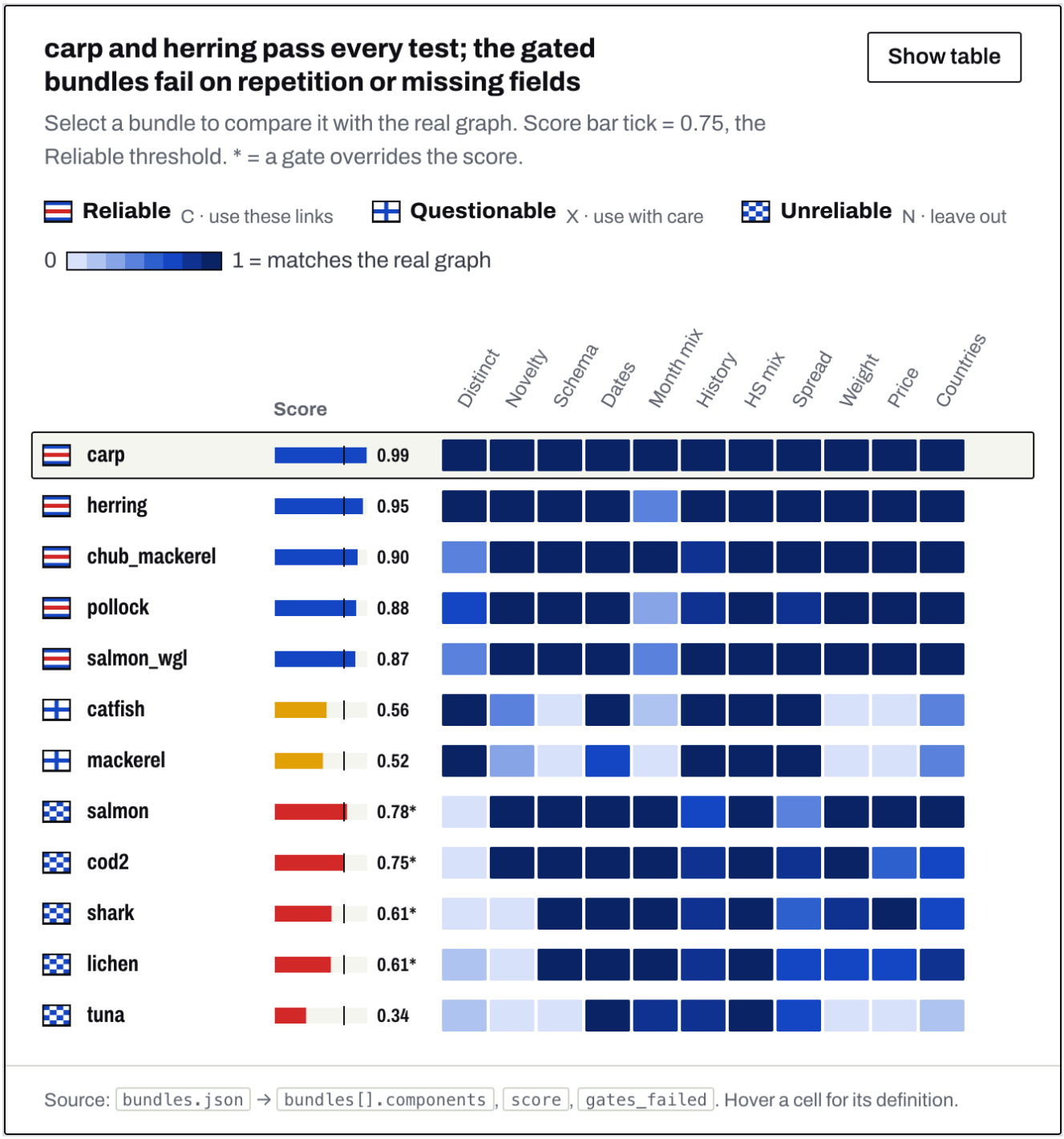


Figure 2.1. Scorecard: carp and herring pass every test; gated bundles fail on repetition or missing fields.

Bundle	Score	Verdict	Decisive evidence
carp	0.99	**Reliable**	165 distinct links, 44% anchored, value bias +0.09 dex, 95% country match (Figure 2.2)
herring	0.95	**Reliable**	clean; month mix slightly off (JSD 0.068 vs null p95 0.026)
chub_mackerel	0.90	**Reliable**	52% of rows repeat, but 155 distinct links pass every realism test
pollock	0.88	**Reliable**	34% repeats, somewhat concentrated (top target 17%)
salmon_wgl	0.87	**Reliable**	57% repeats; 92 distinct links are realistic
catfish	0.56	**Questionable**	no weight at all, 8.7% of dates in 2035, 49% country match, 8% anchored
mackerel	0.52	**Questionable**	the same problems; 32% of dates in 2035
salmon	0.78*	**Unreliable**	132 links are 13 records (9.8% distinct)
cod2	0.75*	**Unreliable**	193 links are **4** records; top target "-53" never receives anything in the graph (Figure 2.4)

Bundle	Score	Verdict	Decisive evidence
shark	0.62	**Unreliable**	19 distinct records, all exact copies of graph edges
lichen	0.61	**Unreliable**	23 distinct records, all copies; value bias +0.26 dex
tuna	0.34	**Unreliable**	no weights, 0% anchored, 18% distinct, 22% country match, 33% of links to one receiver (Figure 2.3)

*A gate overrides the score; on 4–13 distinct records the distribution tests have almost no power.

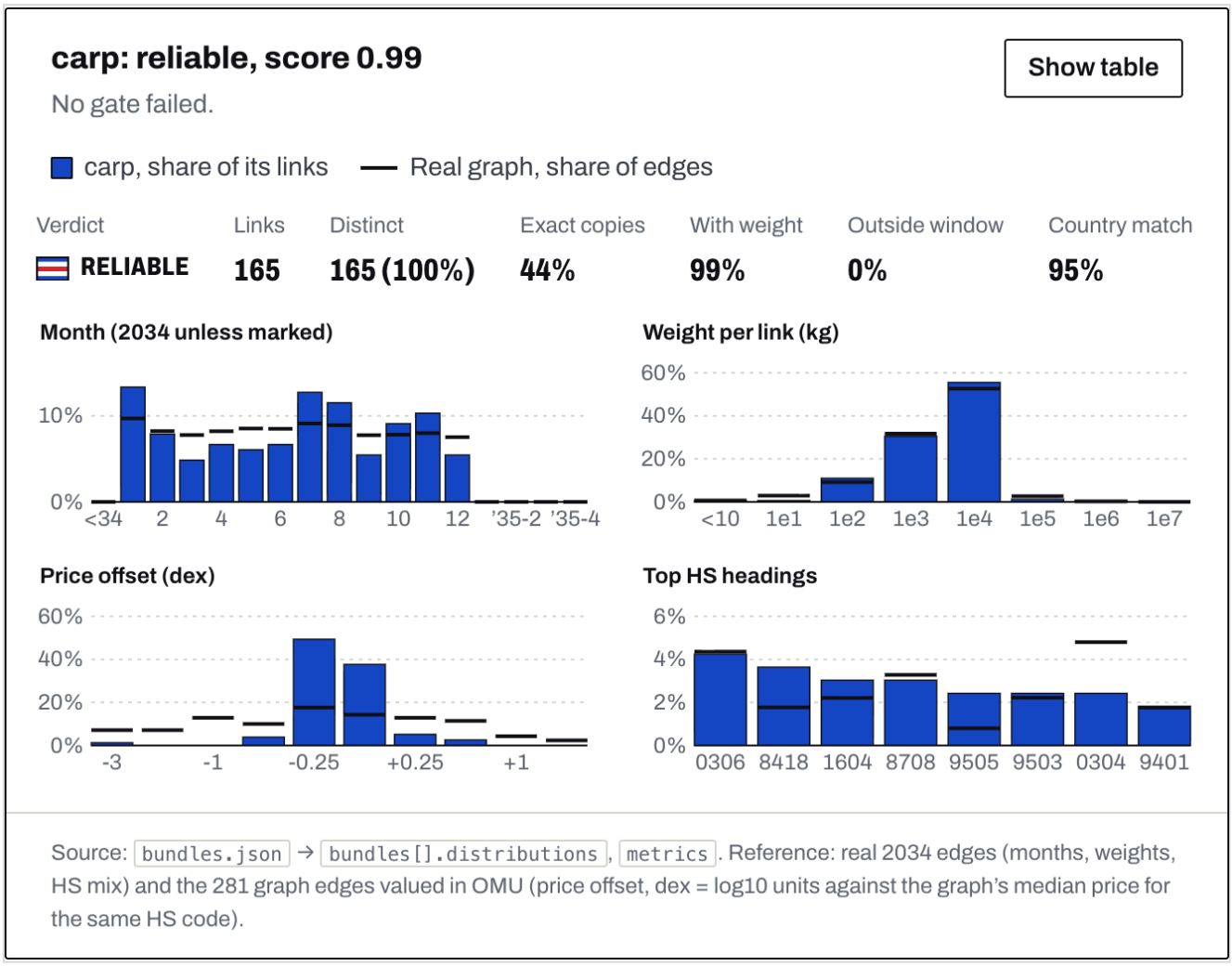
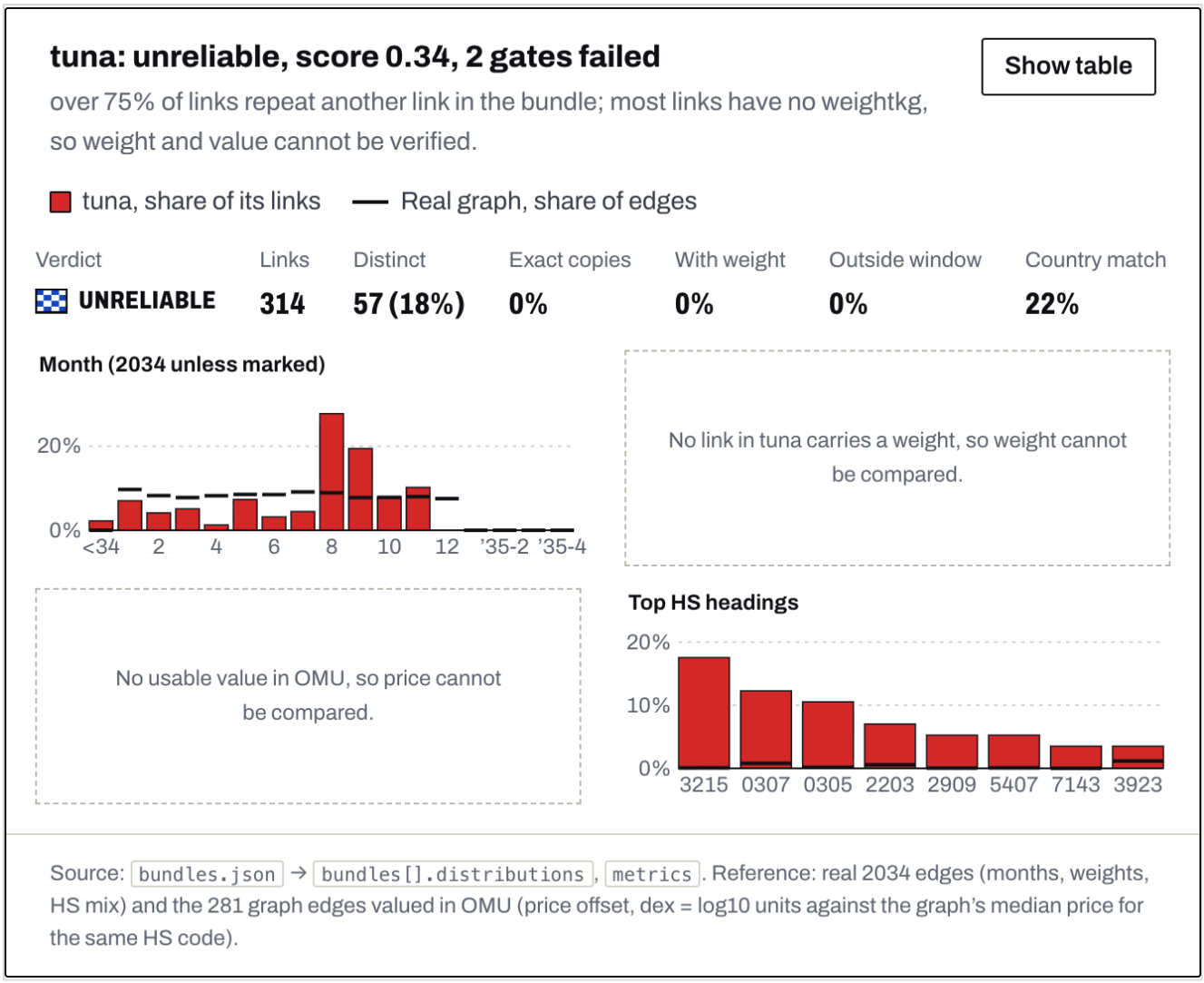


Figure 2.2. carp (score 0.99) against real 2034 edges (black ticks).

Three findings back the verdicts:

1. **Repetition separates the bundles most sharply.** Real data repeats 2.8% of records; the Unreliable bundles repeat 82–98%.
2. **Missing schema identifies the fabricated bundles.** catfish, mackerel and tuna have no weightkg (present on 100% of graph edges). They also disagree on node countries (22–49% vs 93–97%) and rarely match a real record.
3. **Prices.** Reliable bundles' OMU/kg sits within 0.12 dex of the graph's typical price per HS code; lichen and cod2 run high.



Source: `bundles.json` → `bundles[].distributions`, `metrics`. Reference: real 2034 edges (months, weights, HS mix) and the 281 graph edges valued in OMU (price offset, dex = log10 units against the graph's median price for the same HS code).

Figure 2.3. tuna: no weights or values, 18% distinct, a month mix unlike the graph's.

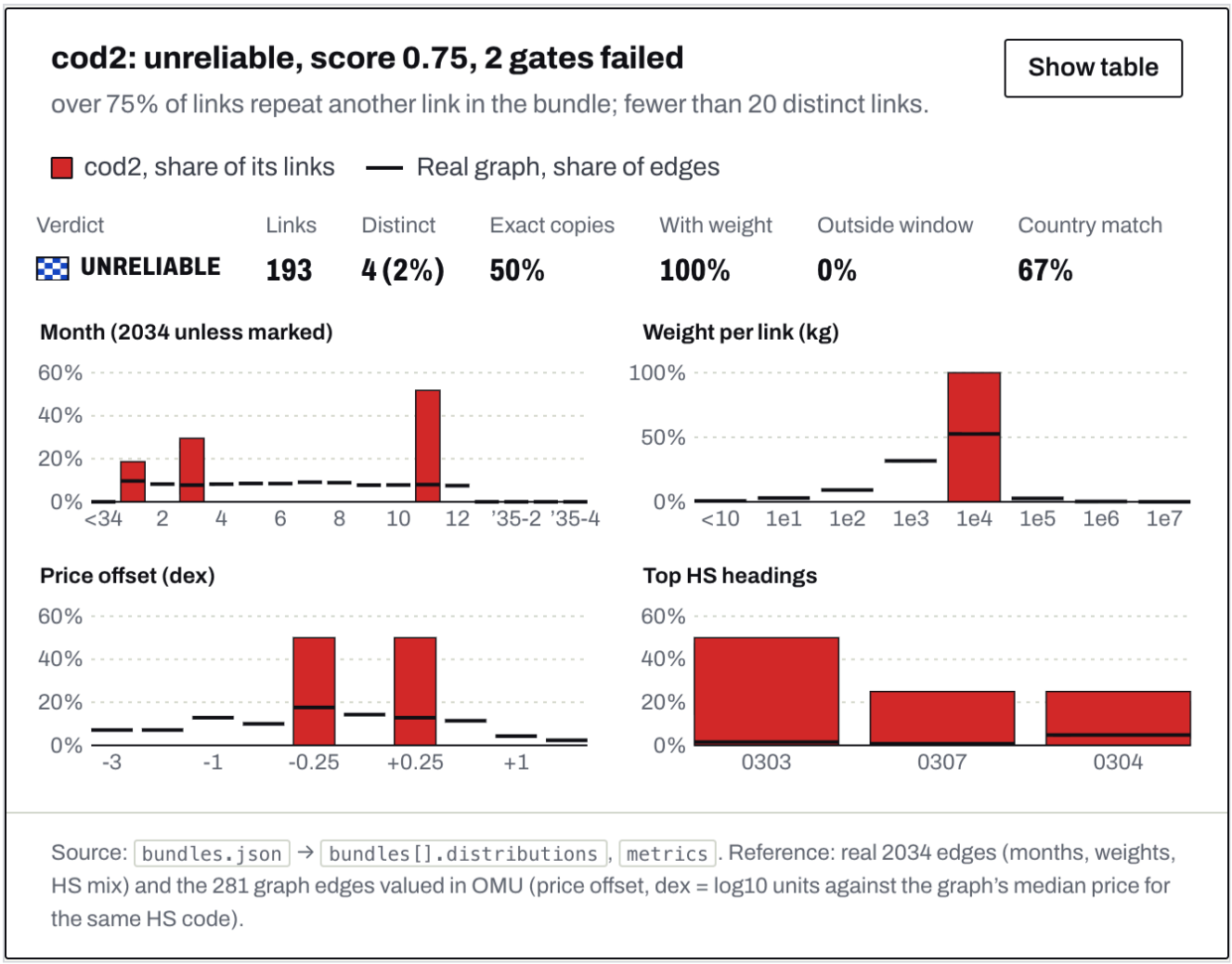


Figure 2.4. cod2: 193 links collapse to 4 distinct records.

The bundles' pair sets barely overlap (Jaccard ≤ 0.05), so corroboration across bundles is impossible.

Most reliable for completing the graph: carp and herring, then chub_mackerel, pollock and salmon_wgl.

Confidence: High for the five Unreliable bundles and for carp and herring. **Medium** for chub_mackerel, pollock and salmon_wgl (realistic distinct links, unexplained repetition) and for catfish and mackerel (weight and value unverifiable).

Question 3

> *Illustrate how your visual analytics approach can be used to identify new patterns and/or anomalies that are present in the knowledge graph after you have added the links you deemed reliable in question 2. Limit your response to 300 words and 4 images.*

- **The links add little (Figure 3.1).** The five Reliable bundles hold 1,015 links; only **347** are usable (distinct, new, inside the window): under 0.1% of 2034 volume. The 2034 fish share stays at 15.6%.
- **Pair patterns barely move (Figure 3.3).** 261 existing pairs gain links and 58 new pairs appear. Only 15 pairs change pattern, mostly one-off or short burst → intermittent, and fading or ramping → steady.
- **Three apparent stops are data gaps (Figure 3.2).** Predicted links keep three "stopped" companies trading: *Chhattisgarh Salt spray OAO United* (receiver, 174 shipments; last real month 2034-03, predicted to 2034-11), *Bahía Azul Ltd Holdings* (shipper, 2034-05 → 2034-12) and *Okavango River S.A. de C.V. Solutions*. Late-2034 stops are partly missing data, so the successor detector requires 6 silent months and marks a pair Unlikely when predicted links revive the predecessor.

- **The flags are stable.** All 68 flagged successor pairs (8 Suspect, 60 Possible) keep their verdicts on the augmented graph, and also after adding the two Questionable bundles (701 links). That robustness is the most useful result.
- **Hubs get richer.** Added links go mostly to existing hubs: *hăi dăn Corporation Wharf* (33), *Sailors and Surfers Incorporated Enterprises* and *Mar del Este CJSC* (11 each). The predictions densify the core rather than reveal hidden companies.

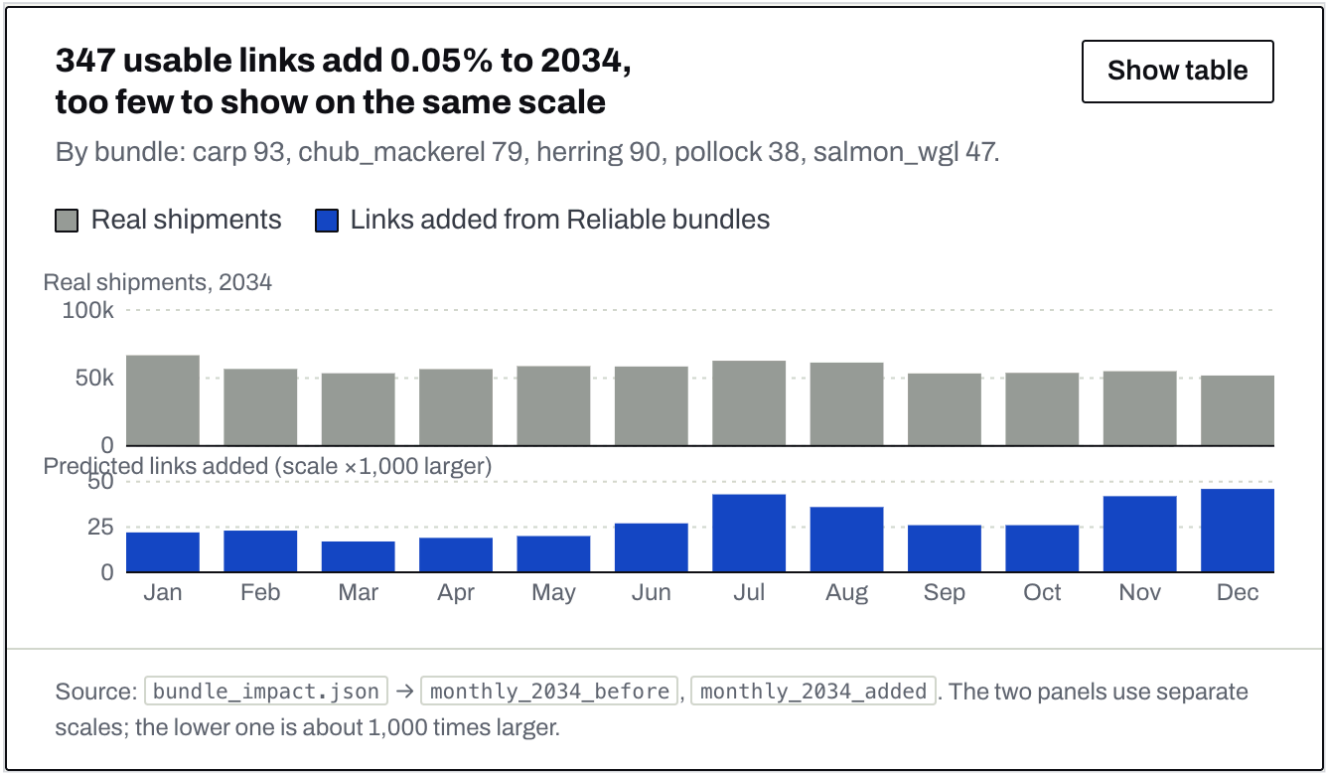


Figure 3.1. 347 usable links add 0.05% to 2034.

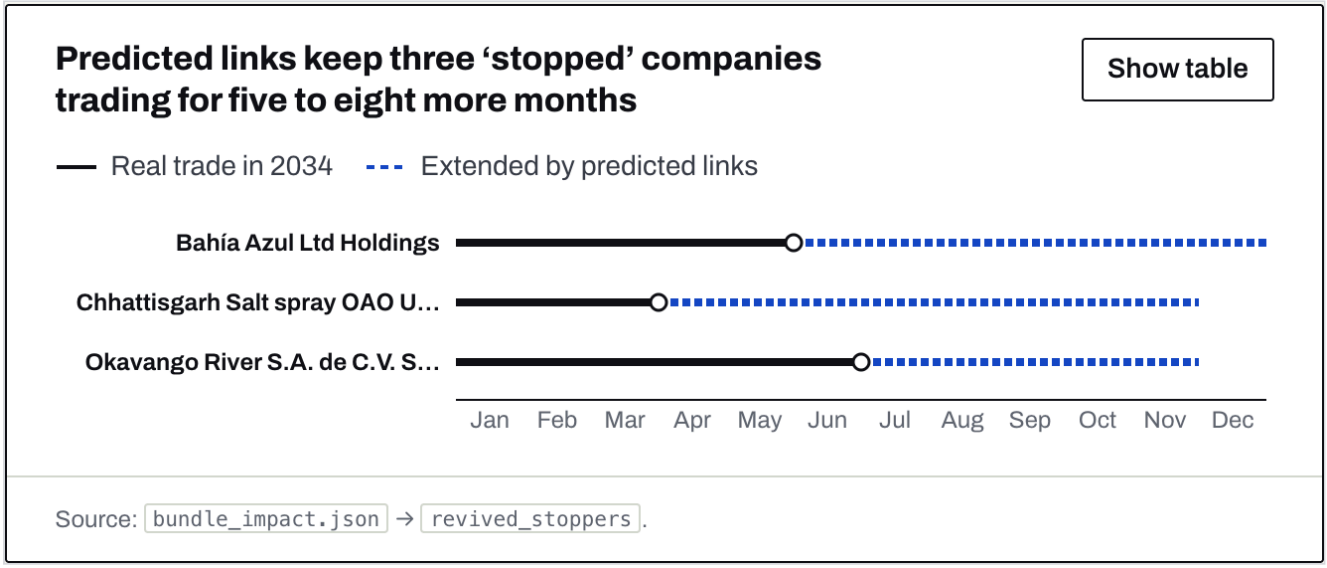


Figure 3.2. Predicted links keep three "stopped" companies trading.

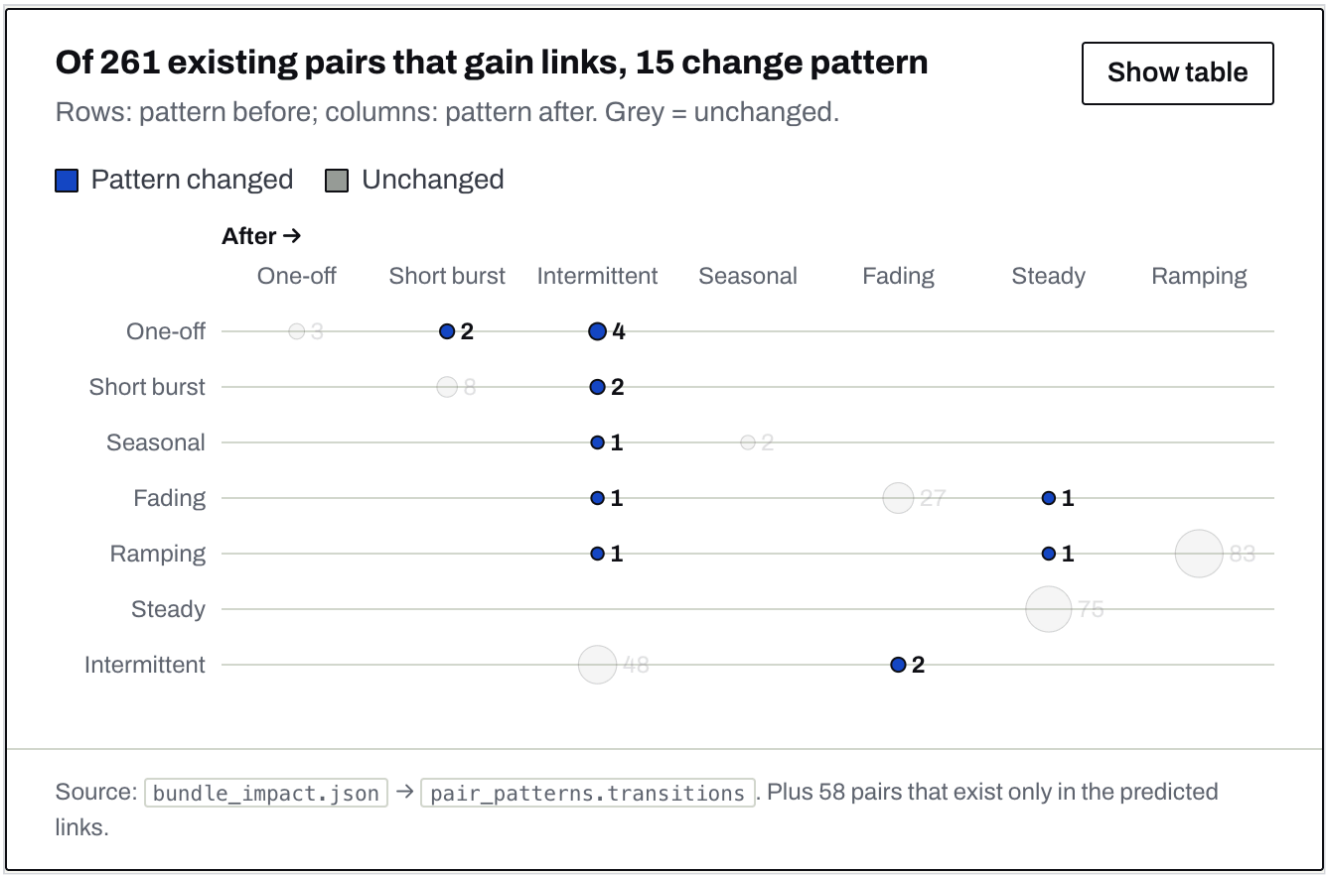


Figure 3.3. Of 261 pairs gaining links, 15 change pattern.

Confidence: High that the impact is small. **Medium** on the revivals: three cases, each from one bundle.

Question 4

> *Identify companies that fit a pattern of illegal fishing. Use visualizations to support your conclusions and your confidence in them. Limit your response to 300 words and 4 images.*

We matched 2,780 shippers and 3,888 receivers that stop ≥ 6 months before the data ends with companies starting −6..+12 months from the stop. Similarity is TF-IDF cosine plus Jaccard on partners and HS codes, so shared hubs count little; shared partners must switch over. Of 872 candidate pairs: 8 Suspect, 60 Possible, 572 Insufficient evidence, 232 Unlikely (Figure 4.2).

Suspect pairs (all trade fish on both sides):

Predecessor → successor	Confidence	Evidence
receiver **-811 → -206**	High	fillets (0304); 2032-12 → 2033-01; 5 shared shippers supply 78% of -206's trade (Figure 4.1)
Beachcomber's Bounty S.A. de C.V. Express → Karelians Limited Liability Company International	Medium	Karelians also succeeds five more fish shippers that stopped in 2030 (Figure 4.3)
Saltwater Surf Club Sp → Chhattisgarh GmbH & Co. KG Sea spray	Medium	dried fish (0305); 2028-12 → 2029-02; all 3 shared partners switch
Asveja Limited Liability Company → Marine Mavericks Marine biology	Medium	molluscs (0307); 95% of successor's trade inherited
receiver **-812 → -556**	Medium	prepared crustaceans (1605); 5 shared shippers
receiver **-513 → -690**	Medium	identical partner set

Predecessor → successor	Confidence	Evidence
The Salty Dog Corporation → Gulf of Guinea Sp	Low	13-shipment predecessor
Estrella del Sol CJSC → Mandalore S.A. de C.V. Logistics	Low	12-shipment predecessor

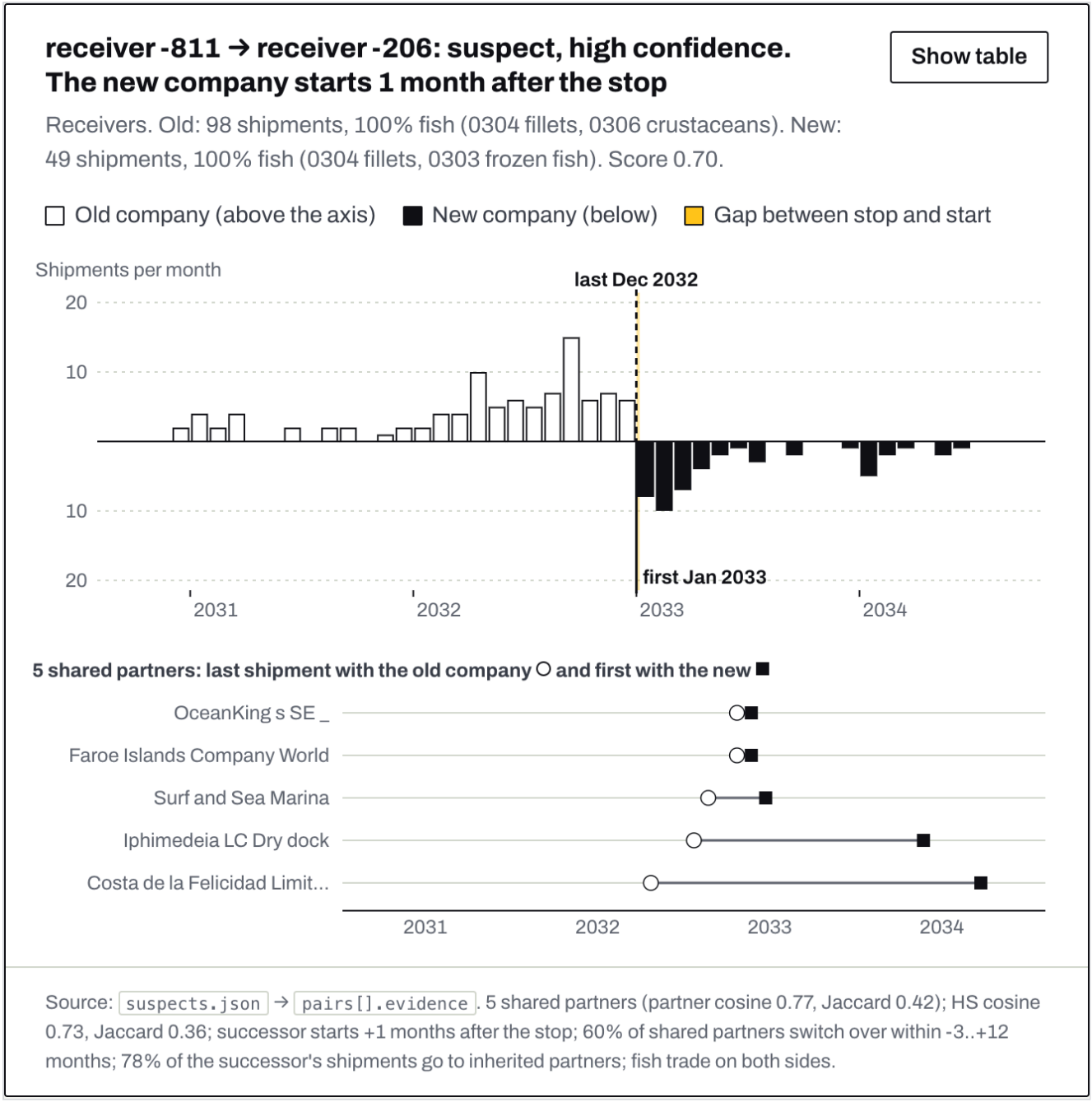


Figure 4.1. -811 stops; -206 starts a month later with its shippers.

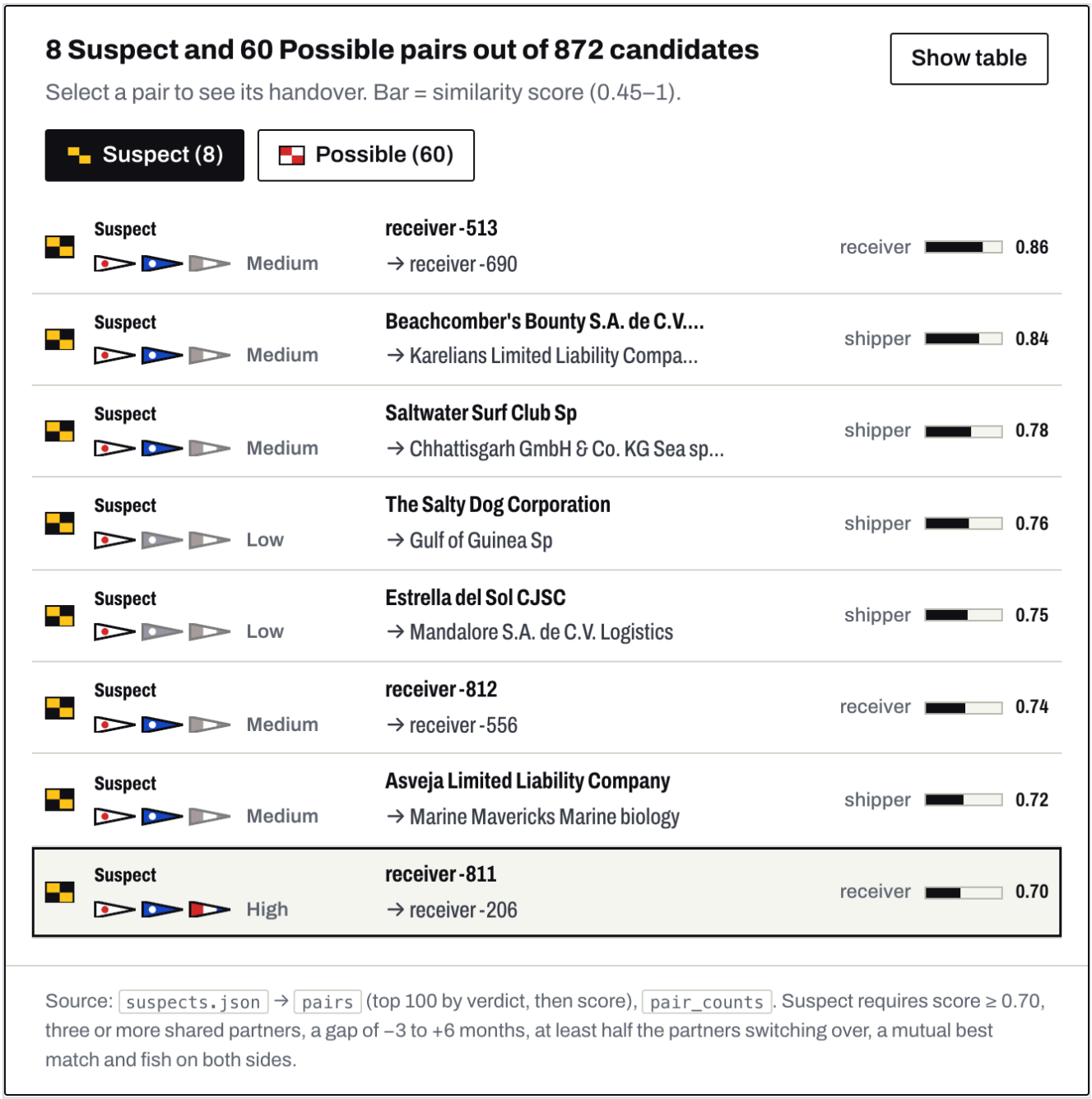


Figure 4.2. 8 Suspect and 60 Possible pairs of 872.

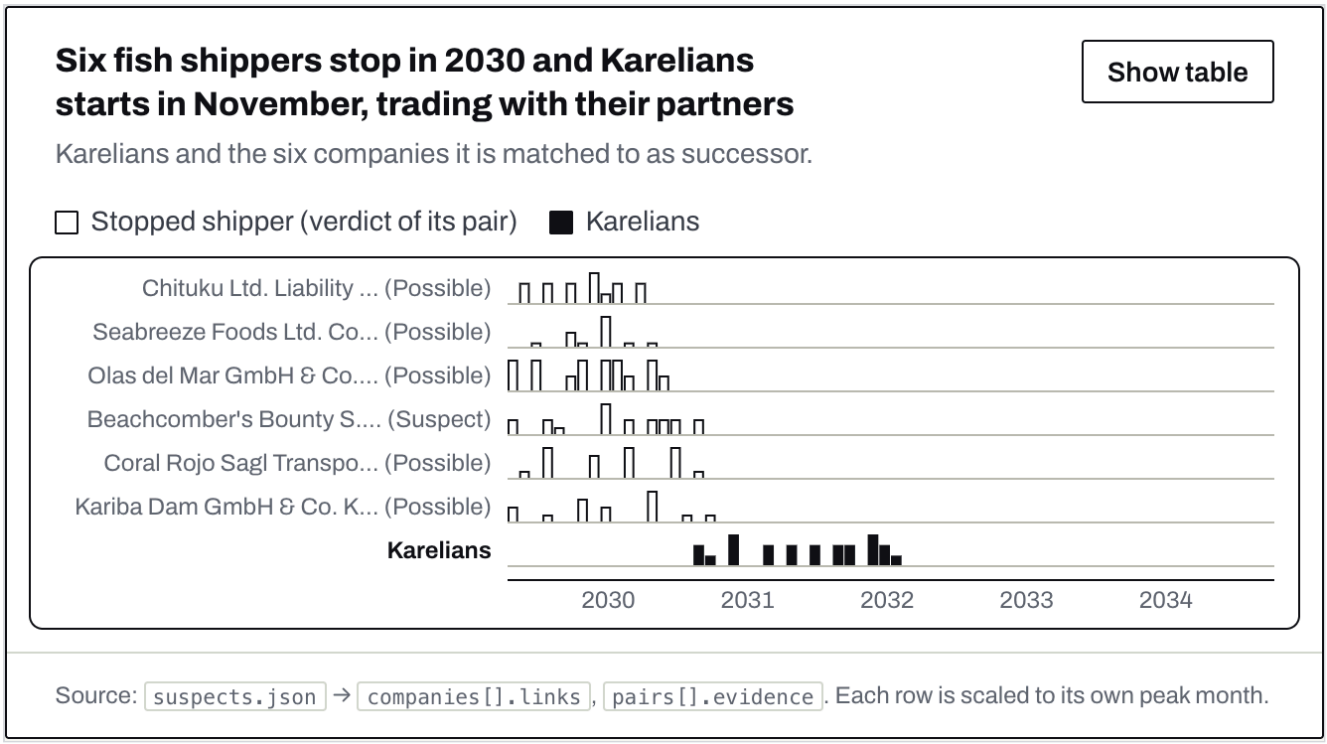


Figure 4.3. Karelians absorbs six stopped fish shippers.

Confidence: Medium overall: only 3–5 shared partners per pair, no ground truth, and 24 high-similarity pairs proved Unlikely because partners did not switch.

Question 5

> *Reflection: What was the most difficult aspect of working with this knowledge graph? Did you have the tools and resources you needed to complete the challenge? What additional resources would have helped you? Limit your response to 300 words*

The data needed the most care.

- **HS codes:** the two code formats would have silently zeroed out chapter-03 fish links in every bundle.
- **Duplicates:** 2.8% of graph edges are exact duplicates.
- **Values:** the graph is in USD and the bundles in OMU, and only 281 graph edges carry OMU, so prices are compared as per-HS-code offsets against those 281 edges.
- **Names:** 22,004 company ids are bare negative numbers, so name similarity is useless as evidence.

Null models changed the conclusions. A first pass scored successors by raw cosine and rated catfish and mackerel "reliable". Weighting partners by rarity and comparing with same-size random samples of real 2034 edges reversed both. The gap profile showed most look-alikes coexist, so similarity alone would flood the list with false positives.

Some evidence is ambiguous. An exact copy of a real edge looked like a defect until five independent bundles each showed 42–59% exact matches; it is better read as evidence that a tool is anchored to real data.

Limitations. Scoring weights and thresholds are judgement calls, published so they can be challenged. Detection needs ≥ 2 shared partners, so very small companies cannot be judged. Stops near the end of 2034 cannot be told from missing data.

Tools and resources. Python with PyArrow (Parquet) made the 5.4M-edge graph tractable, and D3 covered the views. Documentation of the HS-code and currency conventions, and ground truth for a few known restarts, would have helped most. Next steps: compare against the MC1 and MC3 graphs, and let an analyst confirm or reject each pair with scores updating live.

VAST Challenge 2023: Mini-Challenge 3

Entry Name: Signal Hoist

Team Member(s): [Name] — [Affiliation], GitHub @amukherjee1991

Tools Used: Python (pandas, NetworkX, scikit-learn, SciPy, PyArrow), React, D3, Vite, Playwright

Approximately how many hours were spent working on this submission in total? [AUTHOR TO FILL: hours]

May we post your submission in the Visual Analytics Benchmark Repository after VAST Challenge 2023 is complete? [AUTHOR TO FILL: Yes / No]

Video: Not provided

Interactive version: https://amukherjee1991.github.io/Vast23/#/mc3 (every figure below is a screenshot of a view on this page; the views are interactive there)

Questions

Figures are numbered *question.figure*: Figure 1.3 is the third figure in the answer to Question 1. Verdicts use **Suspect**, **Possible**, **Insufficient evidence** and **Unlikely**; confidence is **High**, **Medium** or **Low**.

How the data is modelled. Every link joins a company (source) to a person (target). We resolve the raw records into 16,697 companies, 21,265 persons and 24,931 distinct company–person edges, after splitting 807 links that list several source companies and merging 374 company names with duplicate records (145 of them in more than one country). 8,568 linked company names have no registry record. A *business group* is a connected component: 13,587 of them, with the one giant component split by seeded Louvain into 86 slices. The pipeline is deterministic (byte-identical output across runs).

Question 1

> *Use visual analytics to identify anomalies in the business groups present in the knowledge graph. Limit your response to 400 words and 5 images.*

1. Name collisions create the giant component (Figure 1.1). Persons are identified only by name. Of persons linked to one company, 21.9% have a top-50 surname; of those linked to 5+ companies, 92.6% do (Figure 1.2). The giant holds 7,093 nodes (1,506 companies). Removing the 651 multi-company persons with common surnames shrinks it to 306 nodes; removing as many random multi-company persons leaves 495–1,918 (10 trials). "John Smith", with 12 companies, is likely several people. Confidence: **High**.

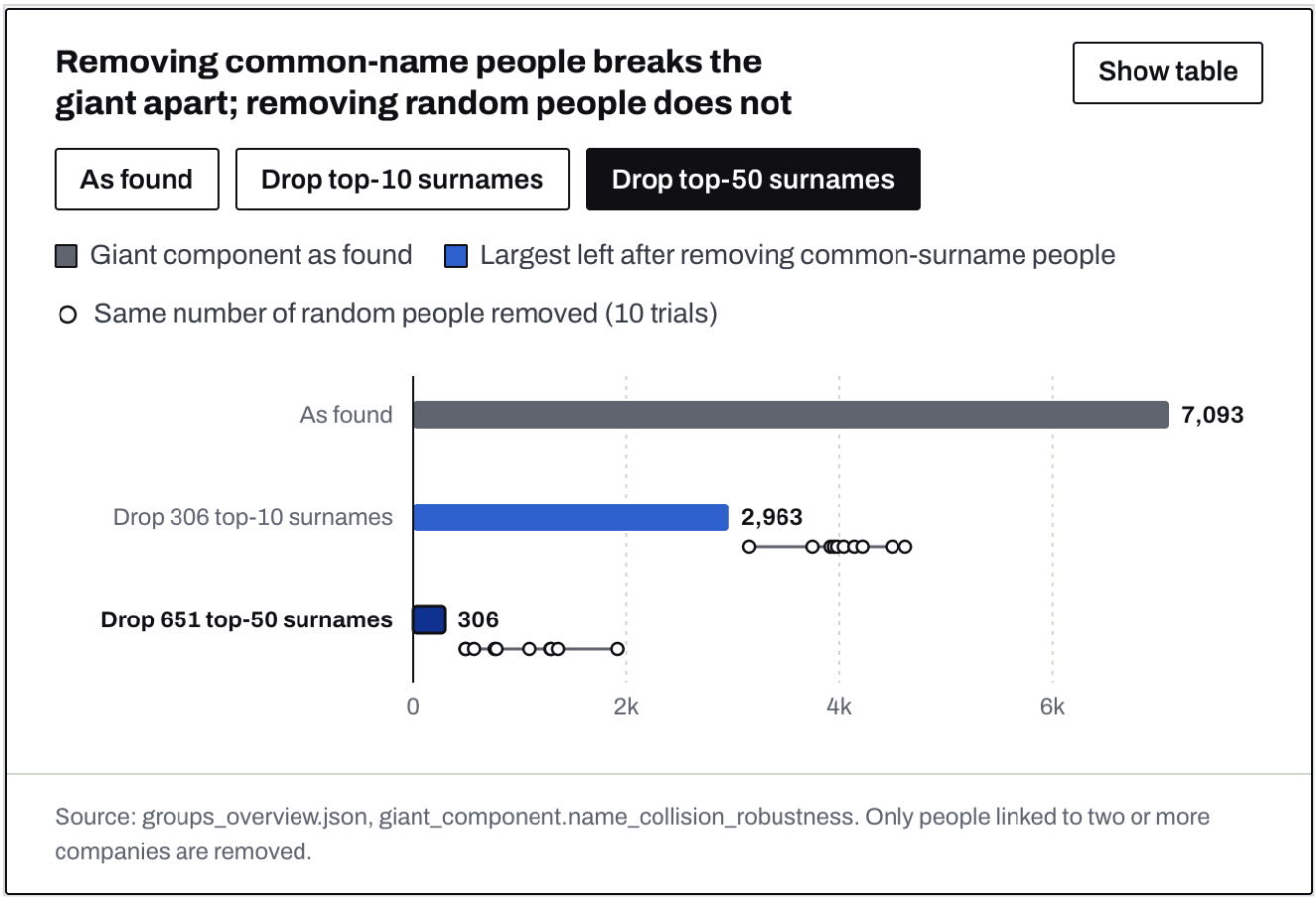


Figure 1.1. Removing common-name people breaks the giant apart; removing random people does not.

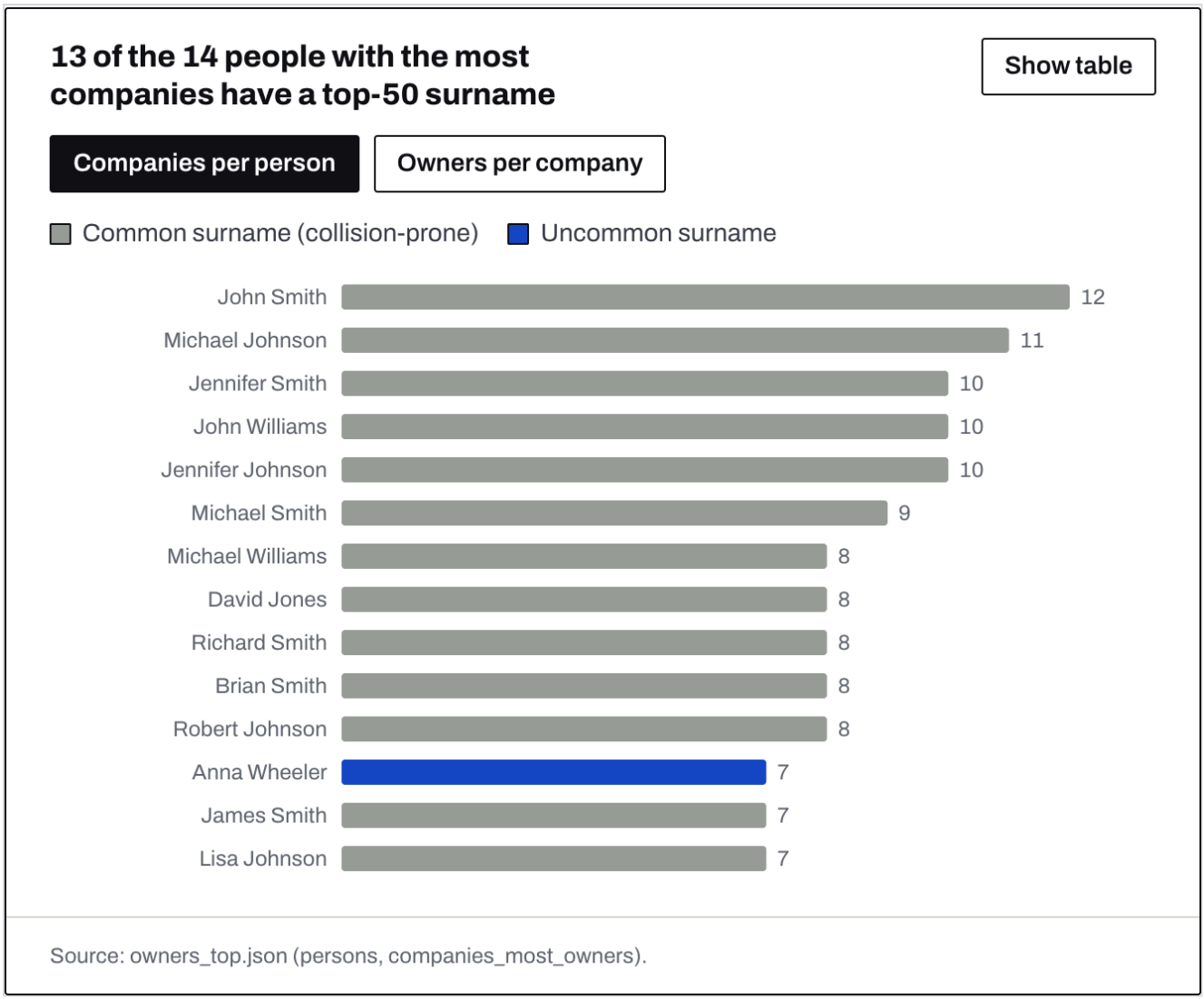


Figure 1.2. 13 of the 14 people with the most companies have a top-50 surname.

2. Ownership fragmentation. Most companies have 0–2 beneficial owners, but 107 have 20+ and 38 have 30+. The largest are *Vespuci Sandbar Sp Brothers* (119), *Dutch Oyster Sagl Cruise ship* (90) and *Niger Bend AS Express* (71). 82 groups carry this flag. Confidence: **High**.

3. Serial owners with uncommon names (least likely to be collisions; Figure 1.4). *Anna Wheeler* owns 7 companies and *Jason Carrillo* is the contact for 6 of them (BG00148). All 7 are unregistered or declare no product. BG00148 has the highest anomaly score of 2,347 scored groups. Confidence: **Medium**: two people sharing an uncommon name remain possible.

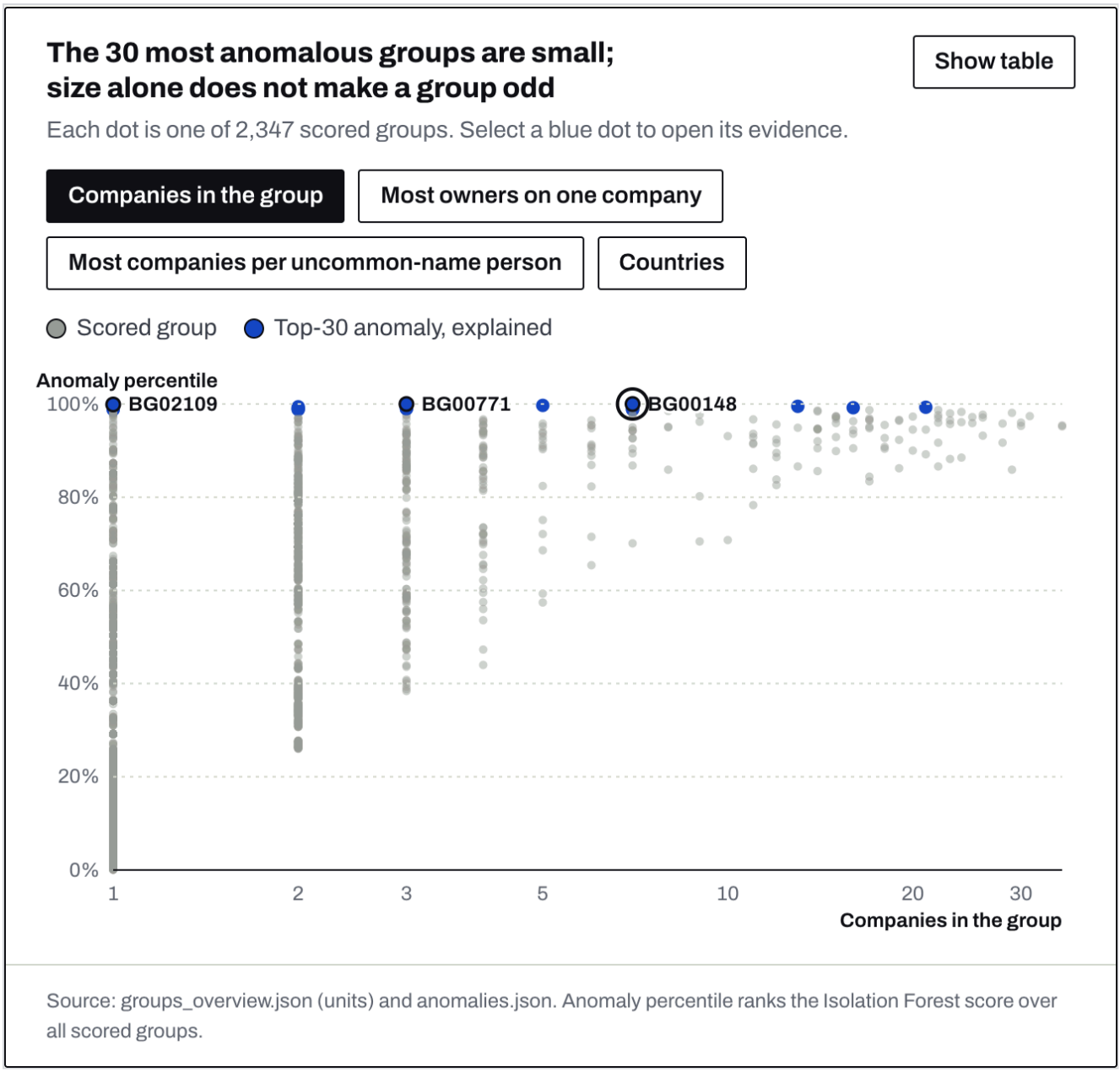


Figure 1.3. The 30 most anomalous groups are small; size alone does not make a group odd.

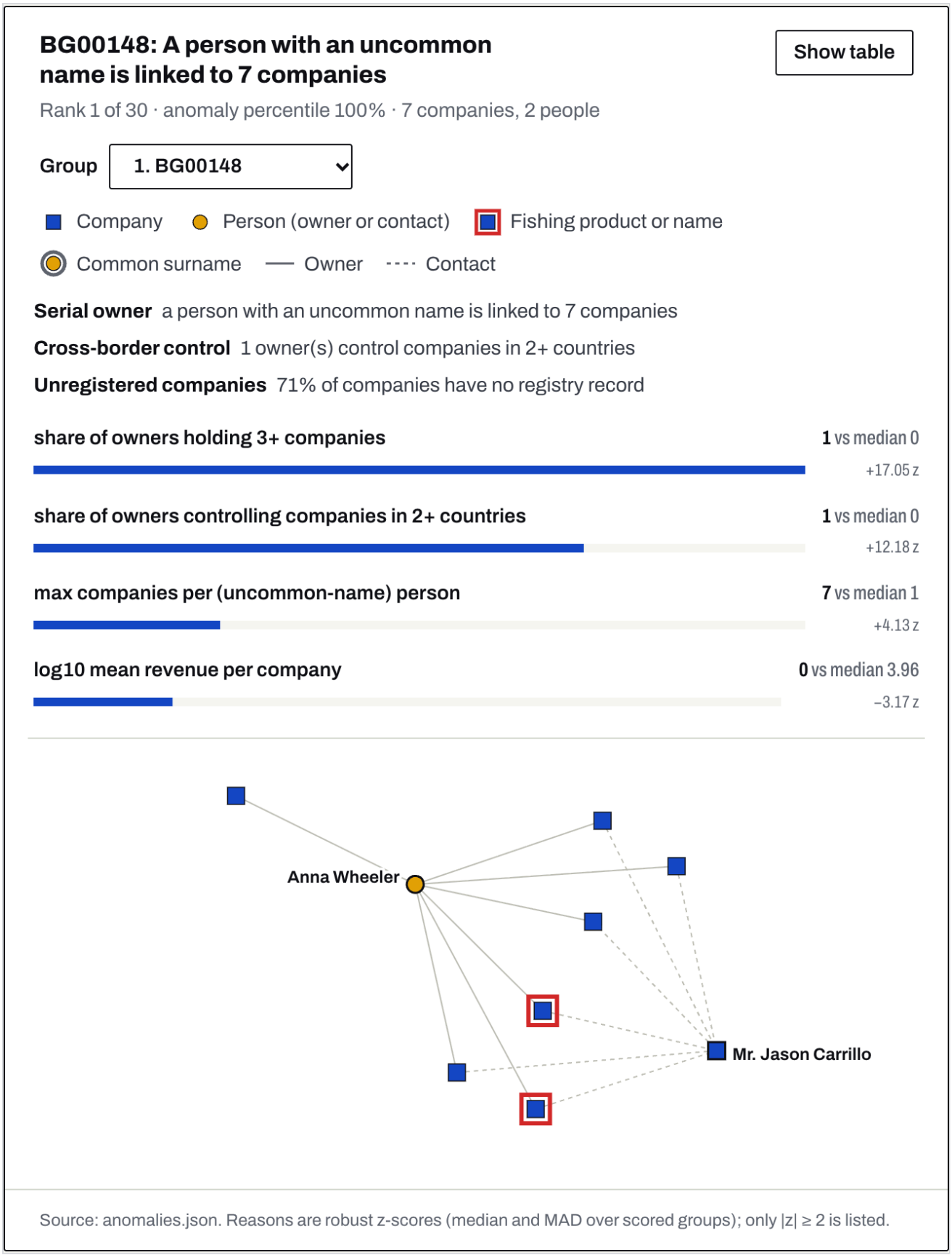


Figure 1.4. BG00148: one uncommon-name person linked to 7 companies.

4. Registry inconsistencies. 112 groups contain a distinctive company name registered in several countries. 166 groups (274 companies) hold a fishing-named firm declaring an unrelated product, such as *Dutch Oyster … Cruise ship*, which sells textiles. *Adriatic Squid* has both "Footwear" and "Seafood" records. Confidence: **Medium**.

5. Data-generation artefacts, not IUU signals: 1,391 names contain a double space where a word was stripped ("Greek ermen"); 113 product texts end in "source: freelance researcher"; 11,420 Faker-style names ("Smith LLC") all sit in country "ZH".

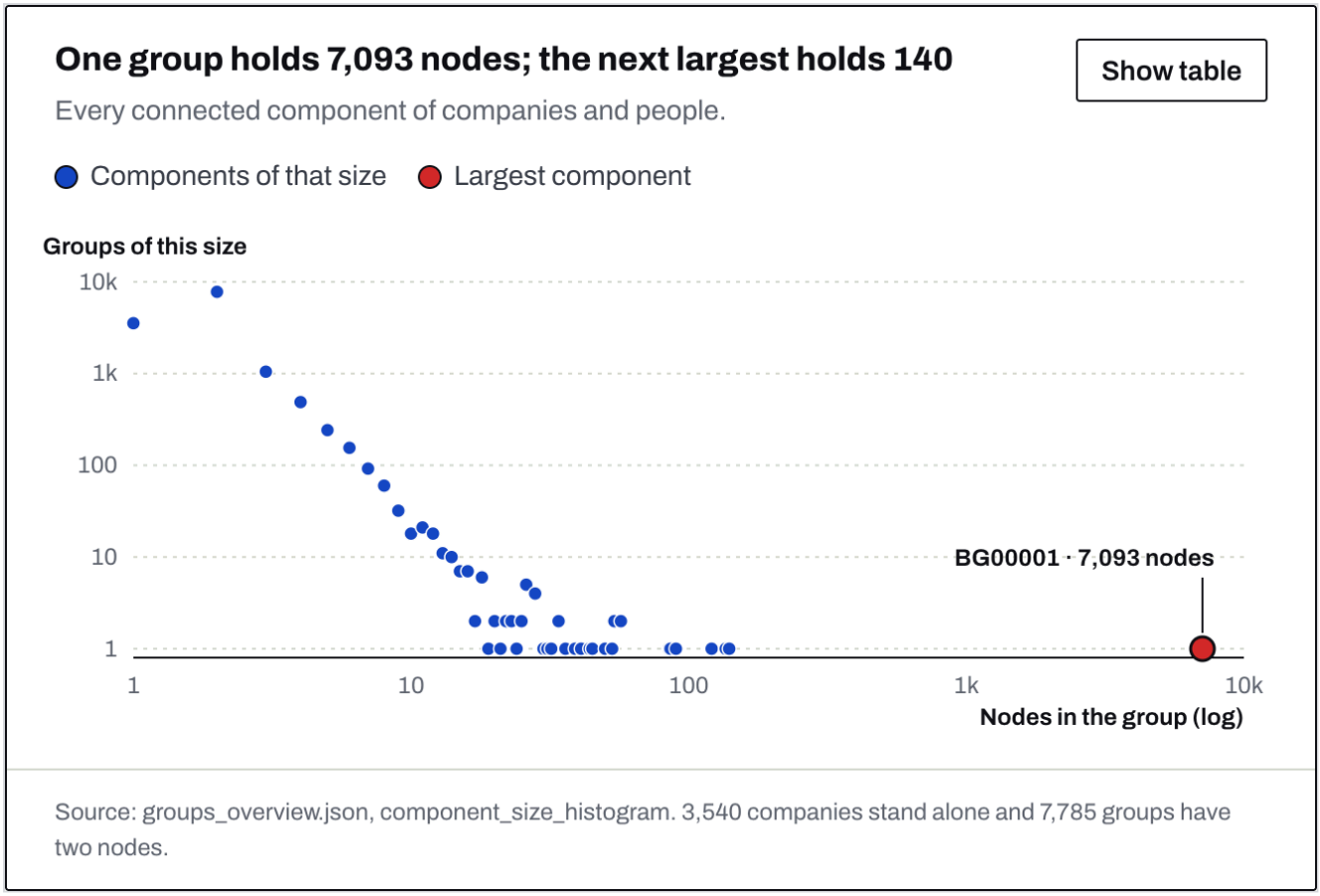


Figure 1.5. One group holds 7,093 nodes; the next largest holds 140.

Method. Each group with ≥ 3 nodes is scored by an Isolation Forest over 16 size-normalised features, explained by robust z-scores (median/MAD), plus 8 rule flags.

Question 2

> *Develop a visual analytics process to find similar businesses and group them. This analysis should focus on a business's most important features and present those features clearly to the user. Limit your response to 400 words and 5 images.*

Process.

1. **Clean and resolve** records by name, as above.
2. **Classify product text** with keyword rules: *fishing* (1,043), *adjacent* (food, cold-chain or shipping; 623), *other* (2,209), *unknown* (12,822). Spot checks look right: "Exotic Coldwater Prawns" is fishing, "Freight forwarding" adjacent, "Gasket" other. Fishing words in company names are a second, independent signal on 1,201 names, including pinyin such as *yú* (fish) and *diao yu* (angling).
3. **Use two similarity spaces.** 77% of companies have no product text; in one shared space the binary "no text" columns dominate the clustering and inflate the silhouette. So:
   - **Text space** (3,875 companies): LSA of TF-IDF on 1–2 word terms (60 dimensions), weight 0.8, plus structural features, weight 0.2 (Figure 2.1).
   - **Structure space** (12,822 companies): structural features only (Figure 2.2).
4. **Choose k** by silhouette × bootstrap stability, because silhouette alone keeps rising with k on sparse text: k = 20 for text, k = 13 for structure (Figure 3.3).

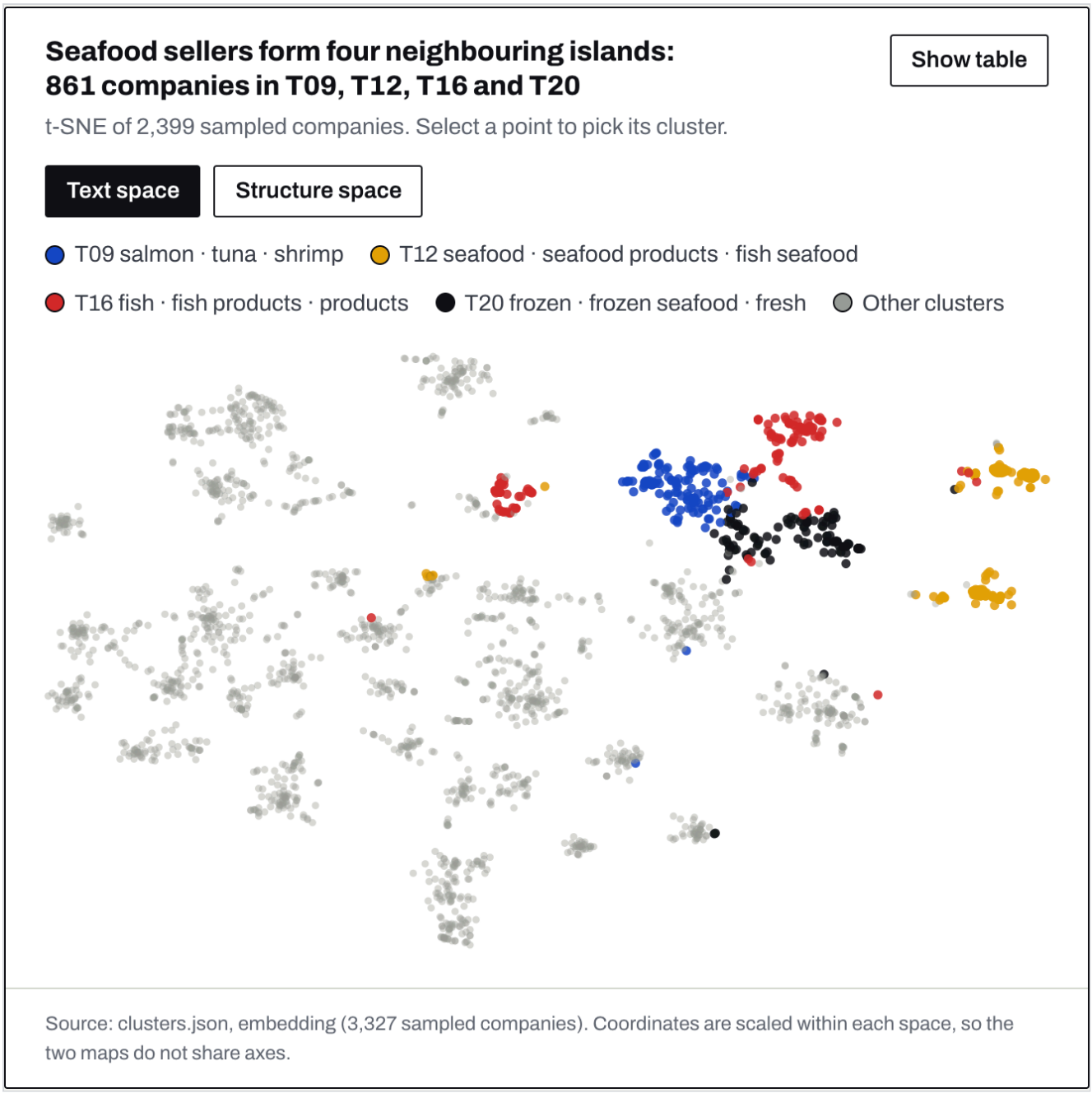


Figure 2.1. Text space: seafood sellers form four neighbouring islands, 861 companies in T09, T12, T16 and T20.



Figure 2.2. Structure space: companies without product text group by registry, ownership and fishing words in names.

Most important features, presented with each cluster's top terms and key features (Figure 2.3):

- **Product terms** drive the text clusters. Four are seafood segments: species (T09: salmon/tuna/shrimp), processed seafood (T12), generic fish products (T16) and frozen seafood (T20). Together they hold 861 companies, each at least 84% fishing-labelled.
- **Ownership structure**: log owners, log contacts and *owner_reuse* (the most other companies held by any of a company's persons). It separates ownership-hub clusters such as S10 from single-contact shells such as S01.
- **Registry status and countries**: unregistered, number of countries and duplicate records give S07 (96 companies with duplicate records in several countries).
- **Fishing or maritime words in names** give S09 and S12, fishing-named firms with no product text; for unregistered companies this is the only fishing signal.
- **Revenue** presence and level separate S03 and S13, high-revenue firms with no product.

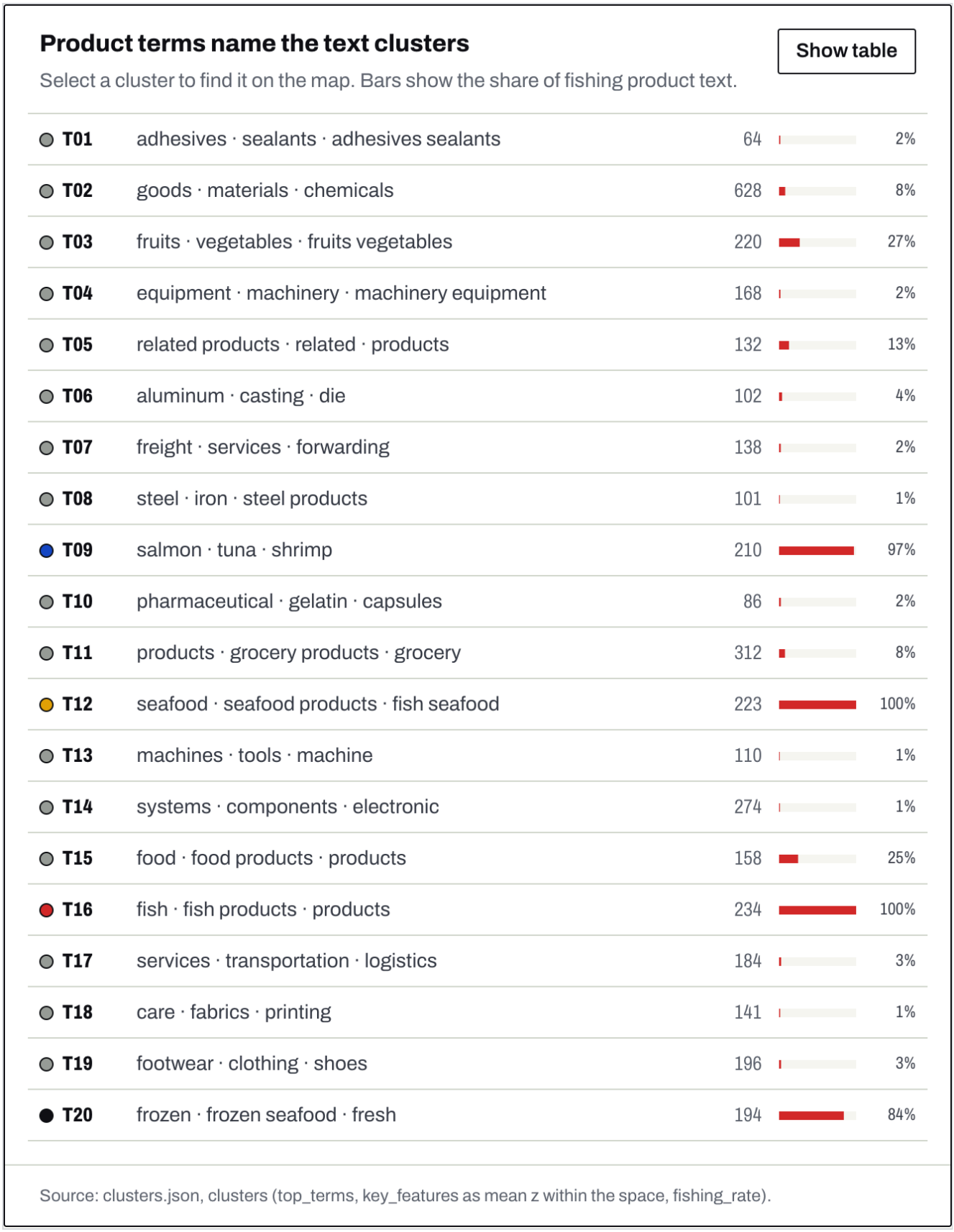


Figure 2.3. Product terms name the 20 text clusters; bars show each cluster's share of fishing product text.

Business groups are compared the same way: each group's 16 robust z-scores form a vector, and cosine similarity finds its nearest groups (listed beside each group in Figure 4.1).

Question 3

> *Measure similarity of businesses that you group in the previous question. Express confidence in your groupings visually. Limit your response to 400 words and 4 images.*

Similarity inside a grouping is measured three independent ways:

1. **Cohesion:** mean pairwise cosine between members, and mean silhouette per cluster.
2. **Stability:** clusterwise Jaccard (Hennig). We refit k-means on ten 80% subsamples and record how well each cluster is recovered.
3. **Membership certainty per company:** margin = $(d_2 - d_1) / d_2$, where d_1 and d_2 are the distances to the nearest and second-nearest centroid; 0 means the company sits on a boundary.

Confidence rule. *High*: stability ≥ 0.85 and silhouette ≥ 0.25. *Medium*: stability ≥ 0.6 and silhouette ≥ 0.05. Otherwise *Low*.

- **Text space:** 2 High, 10 Medium, 8 Low (mean stability 0.60), so the text segmentation is only moderately reliable (Figure 3.1). Processed seafood (T12; silhouette 0.49, stability 0.83) is solid. Generic "fish products" (T16; stability 0.47) blurs into the other seafood clusters.
- **Structure space:** 8 High, 2 Medium, 3 Low (silhouette 0.63, mean stability 0.82). Tighter, but much of that comes from binary features: easy to cluster, not rich.

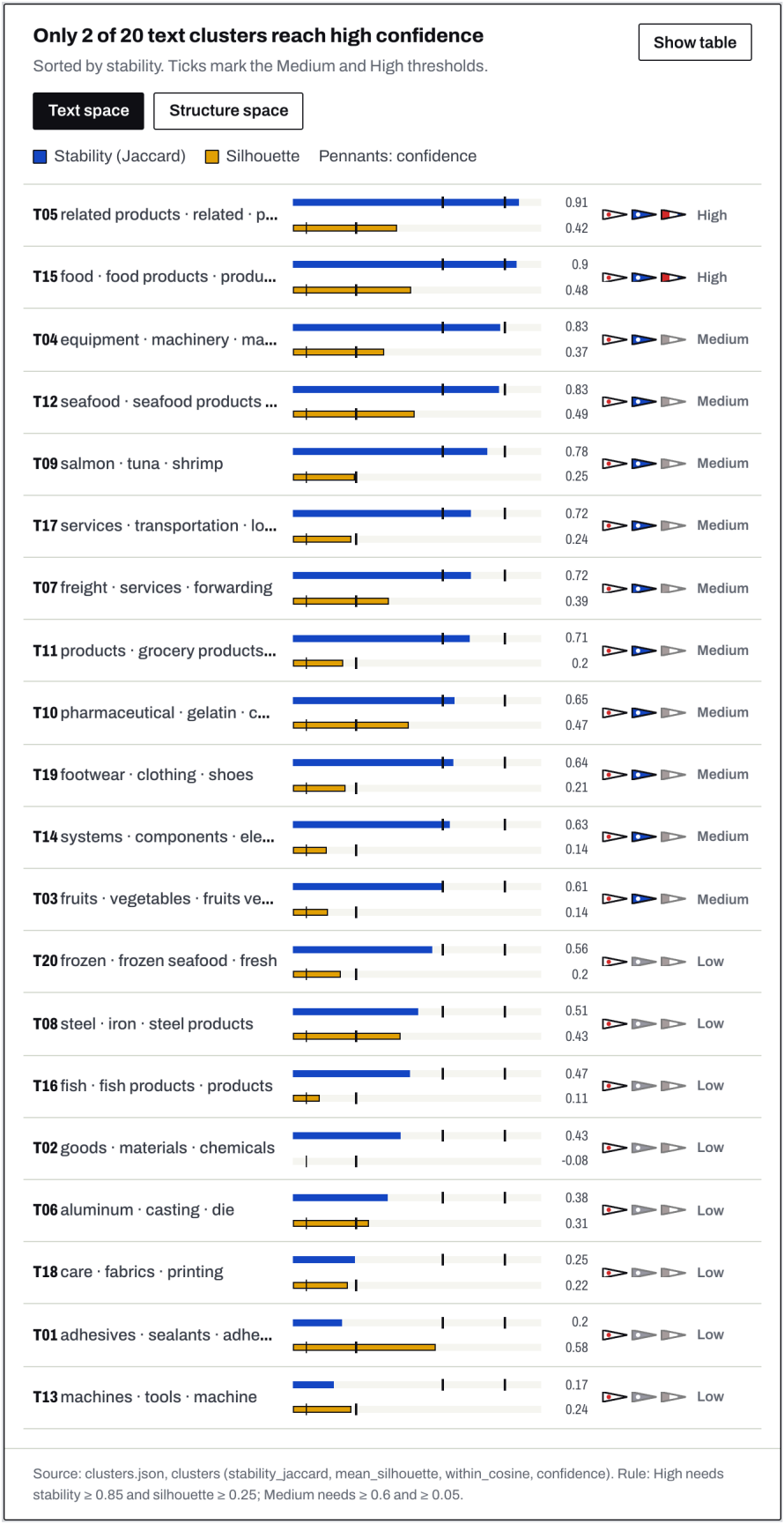


Figure 3.1. Cluster table with bullet bars for stability and silhouette and a confidence badge: only 2 of 20 text clusters reach High.

How confidence is shown visually.

- A t-SNE map per space (3,327 sampled companies), coloured by cluster, with opacity encoding margin so boundary companies fade (Figure 3.2).

- The cluster table with bullet bars for stability and silhouette against the Medium and High thresholds, plus confidence pennants (Figure 3.1).
- A k-selection chart of silhouette, stability and their product, showing why k was chosen (Figure 3.3).
- For business groups, the investigation view shows a confidence badge with the reasons behind it (Figure 4.1).

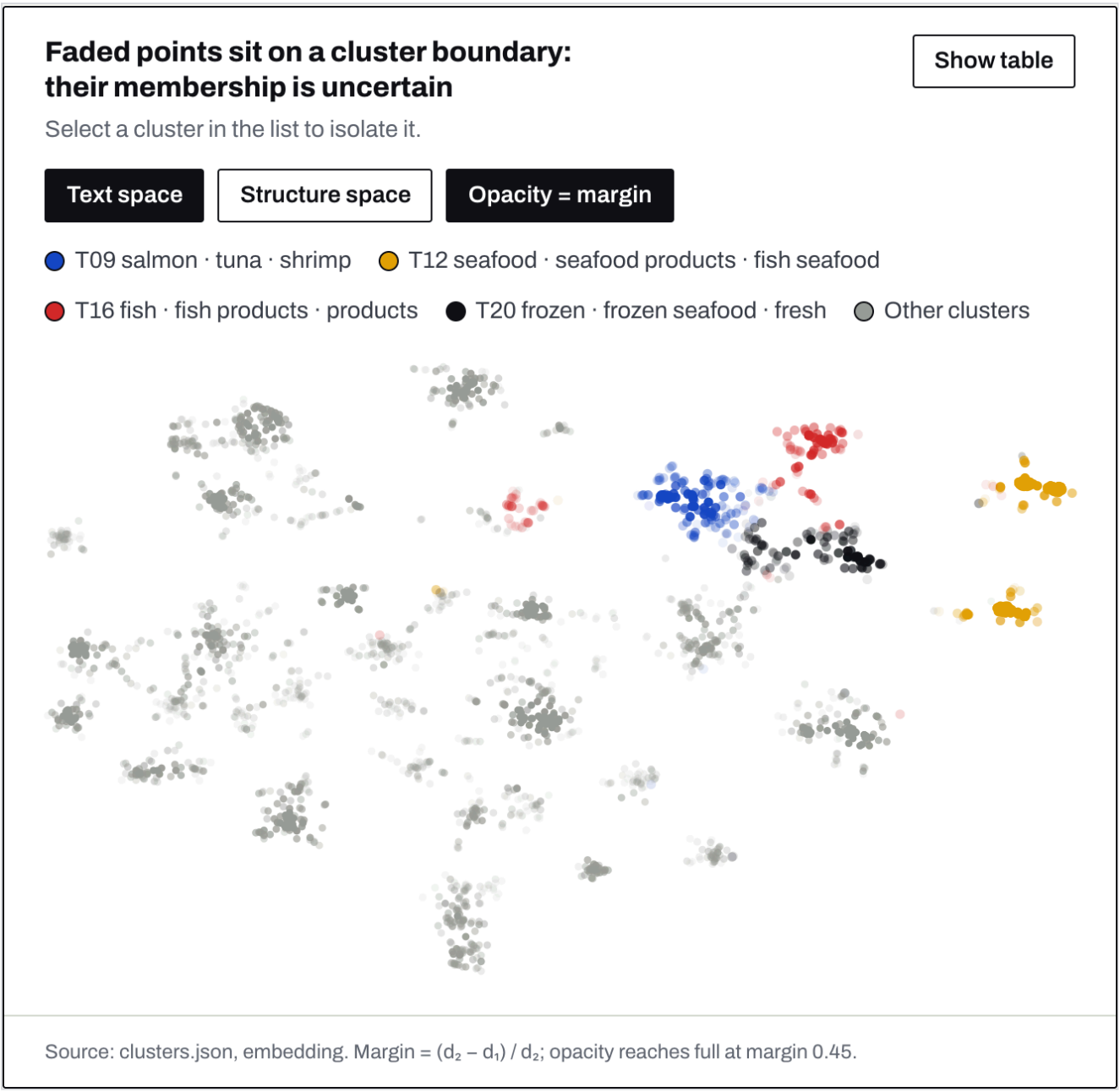


Figure 3.2. Opacity = margin: faded points sit on a cluster boundary, so their membership is uncertain.

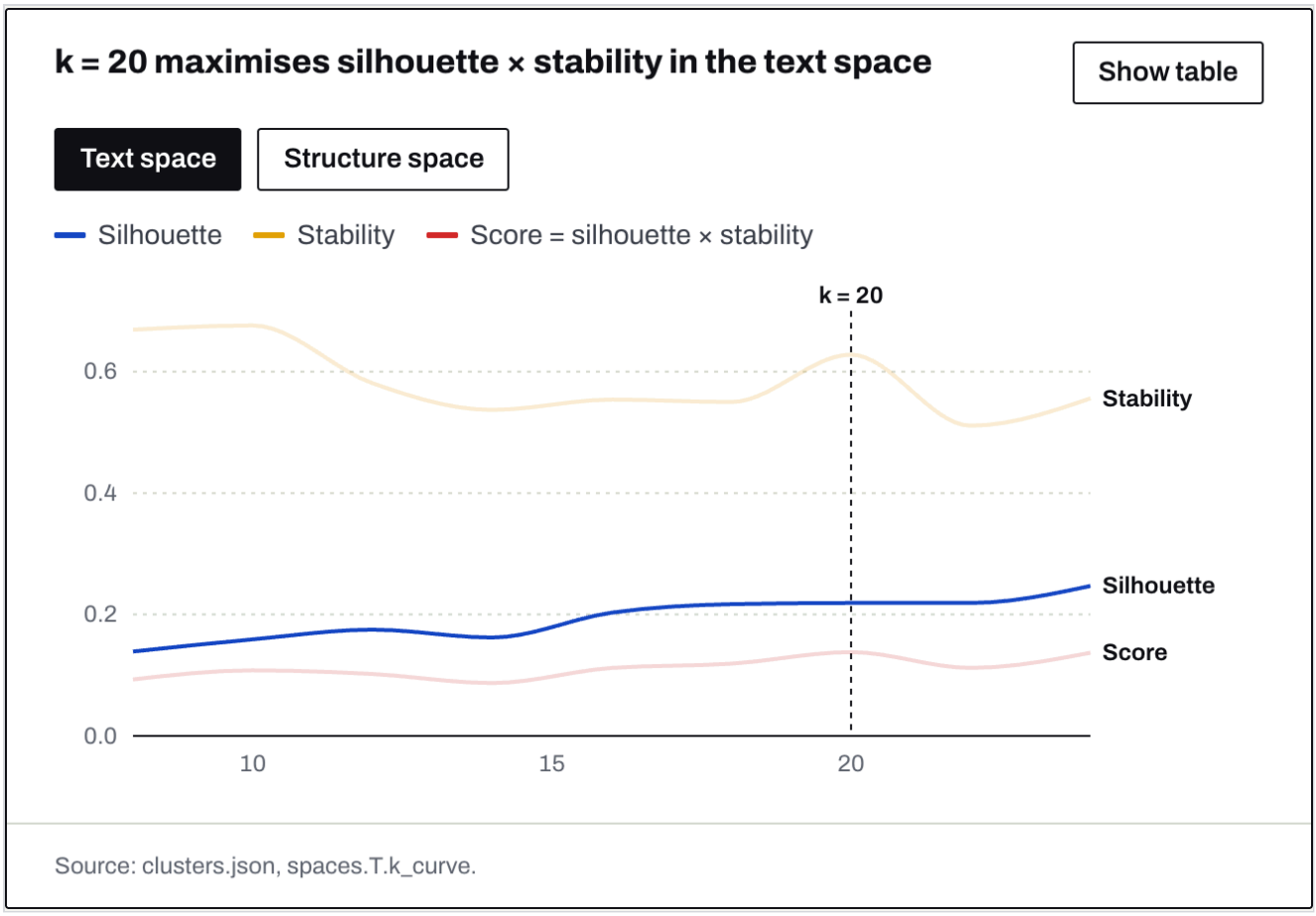


Figure 3.3. k = 20 maximises silhouette × stability in the text space; silhouette alone keeps rising.

Confidence that the clusters are useful for segmentation: **Medium**. Confidence in fine-grained text clusters: **Low**.

Question 4

> *Based on your visualizations, provide evidence for or against the case that anomalous companies are involved in illegal fishing. Which business groups should FishEye investigate further? Limit your response to 600 words and 6 images.*

Short answer. Structural anomaly and fishing activity overlap only moderately. Of the 119 most anomalous groups (top 5%), 13% have fishing relevance of 0.5 or more, against 17% of the rest (Figure 4.2): anomaly alone does not point to fishing. The data cannot prove IUU fishing; it can only rank where to look.

The ranking score is 0.45 × anomaly percentile + 0.30 × flag score + 0.25 × fishing relevance (Figure 4.1). Fishing relevance combines the fishing share of product text with fishing words in names; the product share is shrunk towards the corpus rate of 26.9% (a prior worth 2 companies), so "1 of 1 informative companies is fishing" does not count as 100%. We list 30 groups, with at most 15 Louvain slices of the giant component, each with a rule-based verdict: **2 Suspect, 20 Possible, 7 Insufficient evidence, 1 Unlikely**.

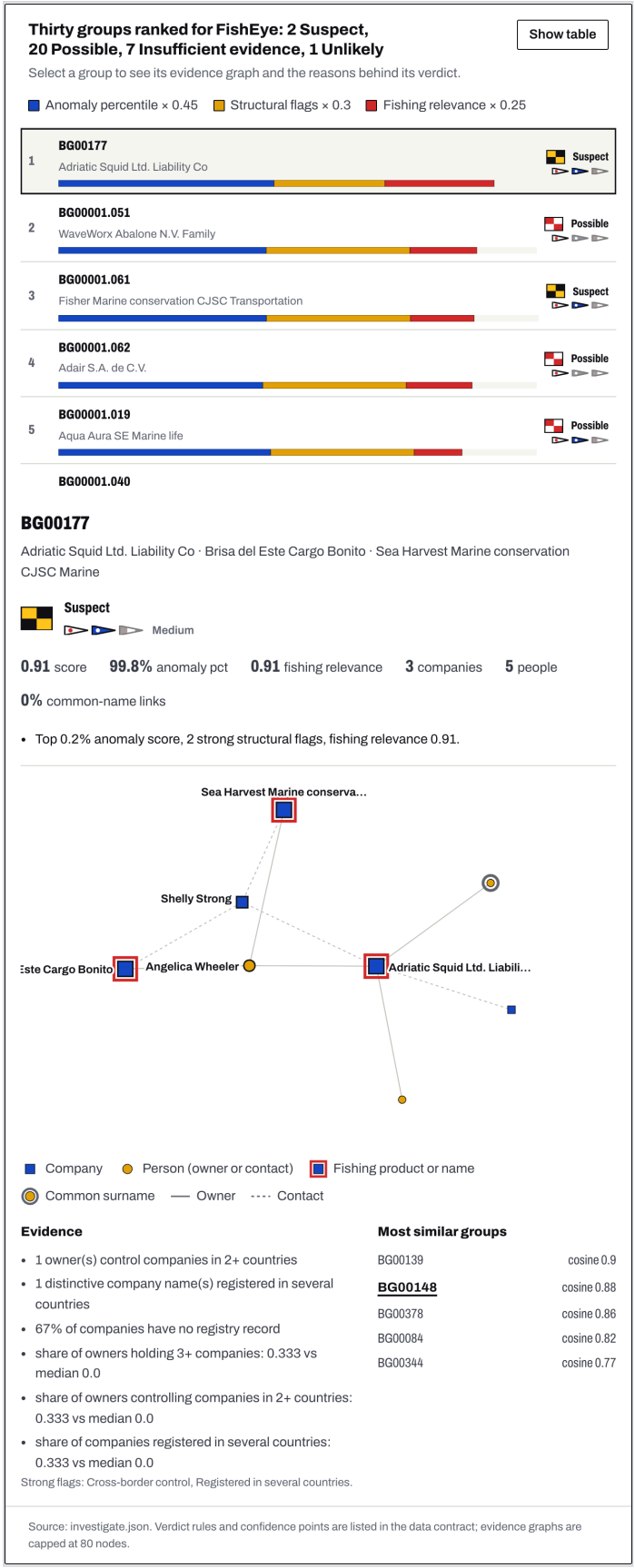


Figure 4.1. Thirty groups ranked for FishEye, with BG00177 open: its evidence graph, reasons and most similar groups.

Investigate first:

1. **BG00177: Suspect, Medium confidence (Figure 4.1).** A whole component of 3 fishing-named companies (*Adriatic Squid*, *Brisa del Este Cargo Bonito*, *Sea Harvest Marine*). One owner (*Angelica Wheeler*) and one contact (*Shelly Strong*) are shared across all three. *Adriatic Squid* is registered in both Marebak and Oceanus, with

conflicting "Footwear" and "Seafood" records. Two of the three are unregistered. Anomaly in the top 0.2% of groups, fishing relevance 0.91, no name-collision exposure.

2. **BG00001.061: Suspect, Medium confidence.** A 16-company slice around *Fisher Marine conservation CJSC* and *Adriatic Tuna Seabass BV*. One company has 26 beneficial owners, a person with an uncommon name links 4 companies, and 62% of the companies are unregistered. As a Louvain slice, its boundary depends on community detection.
3. **BG00148: Possible, Medium confidence (Figure 1.4).** The highest-anomaly group in the data. The *Anna Wheeler* / *Jason Carrillo* pair controls 7 companies across Oceanus and Coralmarica (*Baltic Sprat*, *jīn qiāng yú AG*, *Náutica del Mar*...). None declares a product, so the fishing link comes from names only.
4. **BG00069 and BG00071: Possible, Low confidence.** Small multi-jurisdiction clusters around *Yenisei Eel GmbH* (3 countries) and *Diao yu BV Logistics*. BG00069 would be Suspect, but half of the persons tying each group together have very common surnames.
5. **Single firms registered in several countries:** BG00861 *Diao yu bi sai SRL* and BG00386 *Delhi ers AS Holdings*, both Possible, Medium confidence.

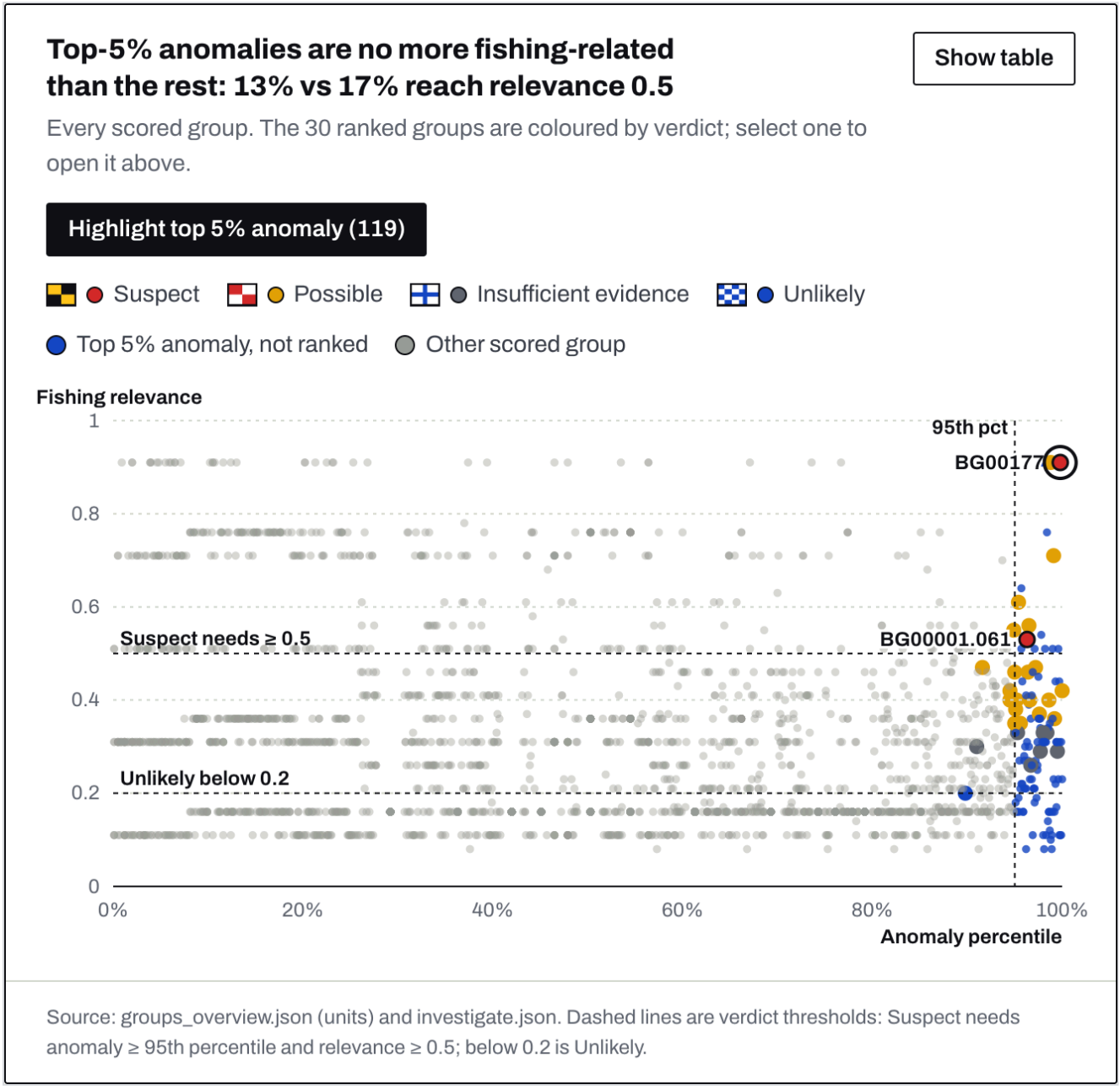


Figure 4.2. Top-5% anomalies are no more fishing-related than the rest: 13% vs 17% reach relevance 0.5.

Evidence against, or where we cannot conclude. Several large groups have strong structure but no verifiable fishing link. BG00003 (23 companies, 113 persons, top company with 27 owners) is *Insufficient evidence* with **High** confidence. The ownership-hub slices (BG00001.001 around *Vespuci Sandbar*, 119 owners) sell wind-power equipment, textiles or beer: they look like shell or nominee structures, but not fishing ones. BG00016 is *Unlikely* (fishing relevance 0.20).

Caveats. 77% of companies have no product text, so most fishing relevance is inferred from names. Person identity is name-only, so collisions make shared control look larger than it is. Weights and thresholds are judgement calls, published with the tool so a reviewer can re-weight them.

Confidence. That the Suspect and Possible groups are a better-than-random IUU lead list: **Medium**. That any single group is engaged in IUU fishing: **Low**.

Question 5

> *Reflection: What was the most difficult aspect of working with this knowledge graph? Did you have the tools and resources you needed to complete the challenge? What additional resources would have helped you? Limit your response to 300 words*

The hardest problem was identity, not anomaly detection. Persons are bare names, so a naïve connected-component analysis produces one 7,093-node "business group", mostly a product of common names such as "Michael Johnson". Testing surname removal against a random control turned that suspicion into a measurement. Every later signal (serial owners, cross-border control) then had to be restricted to uncommon names, and confidence drops wherever a group depends on collision-prone links.

The second lesson was about clustering mixed data. A single feature space reported a reassuring silhouette of 0.49, only because 77% of rows had an all-zero text vector. Splitting text from structure, and choosing k by silhouette × bootstrap stability, gave lower but more honest numbers.

Reproducibility needed explicit work. `Graph.subgraph(set)` iterates in hash order, so Louvain returned 85 or 87 communities depending on the run. Rebuilding every graph with sorted insertion and breaking ties by id made the outputs byte-identical.

Limits. The fishing classifier uses keywords, so it misses euphemisms. Names with stripped words ("Greek ermen") show the generator removed tokens; fragment rules only partly recover them. The verdict rules are transparent but hand-set.

Tools and resources. The open-source Python stack (NetworkX, scikit-learn) and D3 were sufficient. Unique person identifiers, or any attribute beyond a name, would have helped most, along with a hand-labelled sample of fishing companies. With more time we would resolve persons by co-occurrence (two "John Smith"s who share no company are probably different people), validate the fishing labels against that sample, and test how sensitive the investigation ranking is to its weights.